

IBM Z AIOps

Simplifying Mainframe Operations

John Baker

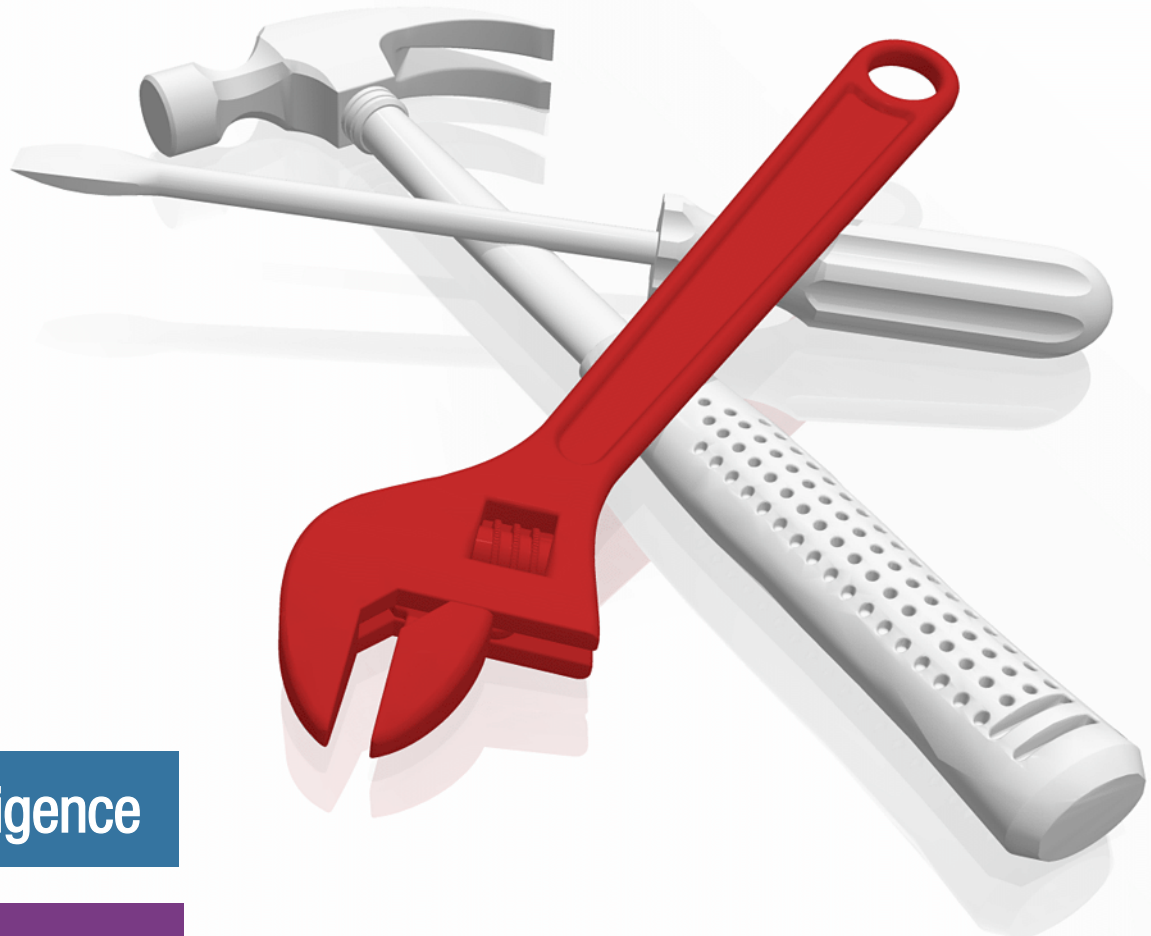
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Artificial Intelligence

IBM Z



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IBM Z AI Ops: Simplifying Mainframe Operations

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Note: Before using this information and the product it supports, read the information in “Notices” on page v.

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Contents

Notices	v
Trademarks	vi
Preface	vii
Authors	viii
Now you can become a published author, too!	ix
Comments welcome	x
Stay connected to IBM Redbooks	x
Chapter 1. Introduction	1
1.1 Driving forces behind modernizing the IBM Z platform	2
Chapter 2. Challenges	3
2.1 Too much data and siloed organizations	4
2.2 Lack of observability	5
2.3 How to optimize for availability and performance	6
2.4 Increased focus on compliance and vulnerabilities	7
2.5 Cost and speed	7
2.6 Mainframe skill gap challenges	8
Chapter 3. Artificial Intelligence for IT Operations capabilities	9
3.1 Industry background: IT service management and ITOps	11
3.1.1 IT service management	11
3.1.2 ITOps	11
3.1.3 Incident management	12
3.2 Observability and monitoring	16
3.2.1 Defining observability	16
3.2.2 Defining monitoring	16
3.2.3 Establishing visibility into z/OS	17
3.3 Performance and capacity management	21
3.4 Automation	23
3.4.1 Overview	23
3.4.2 History	25
3.4.3 Automated operations	25
3.4.4 Batch processing or job scheduling	28
3.5 The role of artificial intelligence in ITOps	31
3.5.1 Brief history and evolution	32
3.5.2 Anomaly detection	32
3.5.3 Generative AI	34
Chapter 4. Simplifying mainframe operations	37
4.1 Single point of control for IT operations	38
4.1.1 Operations management	38
4.1.2 Resiliency and risk management	44
4.1.3 Security and compliance management	45
4.2 Observability: Mainframe scoped and across the enterprise	45
4.2.1 Telemetry aggregation and distribution	45
4.2.2 Visibility into transaction traces	46
4.2.3 Augmenting transaction traces with infrastructure metrics	48

4.2.4 Adopting OpenTelemetry	48
4.3 Adopting AI for mainframe operations.	49
4.3.1 Smart alerts.	49
4.3.2 Event correlation.	49
4.3.3 Predictive insights.	50
4.3.4 Short interval data correlation	50
4.3.5 Faster root cause analysis	51
4.4 Optimization and availability	52
4.4.1 Performance and capacity	52
4.4.2 Automation: better together	55
4.4.3 Workload Automation	59
Chapter 5. Where to start	63
5.1 Establishing a solid monitoring foundation	64
5.2 Developing a strategy for performance and capacity planning	64
5.3 Establishing standards for automation	65
5.4 Telemetry distribution	66
5.5 Using the unified dashboard for operations	66
5.6 Further advanced steps	67
5.6.1 Establishing an observability strategy.	67
5.6.2 Improving asset management.	67
5.6.3 Backup management	68
Chapter 6. Integration architecture	69
Chapter 7. Conclusion	71
Abbreviations and acronyms	73

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Preface

Many of the world's transactional workloads run on the mainframe. Organizations face a critical challenge: helping ensure the availability of workloads and an infrastructure that supports essential business applications. Many enterprises report two major challenges: an overwhelming volume of data to process and too many tools with disjointed user experiences.

The mainframe does not operate in isolation. You must navigate integration complexity between IBM Z and distributed cloud environments. This complexity often prevents end-to-end observability and makes it difficult to meet service-level goals.

This IBM Redbooks® publication addresses a common question: How can you simplify mainframe operations? To answer this question, this IBM Redbooks publication draws on Lean Thinking, which focuses on identifying waste (muda) and strain (muri) in any process. Waste includes manual data gathering, redundant steps, and fragmented tools. Strain refers to the difficulty of identifying root causes from large volumes of data.

By applying artificial intelligence (AI) and machine learning (ML) technologies to mainframe operations, you can improve efficiency (reduce muda) and effectiveness (reduce muri).

For example, AI-powered solutions detect operational anomalies, correlate them, and guide you with expert advice to take the next best action, without requiring you to sift through multiple tools and datasets. Proactive performance analysis replaces reactive troubleshooting, helping you address issues before they affect business. With enhanced visualization, open standards such as OpenTelemetry, and real-time system health insights, you gain actionable intelligence for both immediate decisions and long-term planning.

Generative AI-based assistants and agents streamline and automate tasks, transforming how you interact with the mainframe. Natural language interfaces simplify complex processes, making operations more intuitive and productive for users at all experience levels. This modernized approach improves efficiency and reliability across your environment. This IBM Redbooks publication provides best practices to help you simplify mainframe operations.

If you are a mainframe leader or CTO, this IBM Redbooks publication provides a comprehensive overview of the Artificial Intelligence for IT Operations (AIOps) landscape on the mainframe, required capabilities, and available solutions. You learn how these solutions work together to deliver secure, robust, efficient, and simplified operations, despite the challenges mainframe organizations face today. To explore these challenges and key capabilities, see Chapter 2, “Challenges” on page 3 and Chapter 5, “Where to start” on page 63 through Chapter 7, “Conclusion” on page 71.

If you are an IT operator or subject matter expert (SME), this IBM Redbooks publication helps you understand the diverse capabilities that are required in IT Operations (ITOps). You explore key operational areas and learn why each capability matters. Chapter 3, “Artificial Intelligence for IT Operations capabilities” on page 9 and Chapter 4, “Simplifying mainframe operations” on page 37 provide deeper insights into monitoring and observability, performance and capacity planning, automation, and how AI transforms your AIOps experience.

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¹ Sean Gillespie, et al, *Lean for Sales: Bringing the Science of Lean to the Art of Selling*, 2016, Taylor & Francis, ISBN 9781482253146

² Bidanda, *Maynard's Industrial and Systems Engineering Handbook, 6th Edition*, 2023, McGraw Hill, ISBN 9781260461565

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Introduction

Artificial Intelligence for IT Operations (AIOps) is a broad term that includes tools and capabilities that are required to operate an enterprise IT infrastructure and its applications. This IBM Redbooks publication provides an orientation to AIOps in a heterogeneous, hybrid IT environment that can include public and private clouds, with a specific focus on IBM Z.

IBM Z differs from distributed and cloud environments because of its unique characteristics, developed over six decades of innovation. The mainframe platform offers instrumentation and operational best practices that are unmatched by distributed systems. Key features include the following functions:

- ▶ Comprehensive instrumentation and operational best practices
- ▶ 99.999999% availability¹
- ▶ Exceptional transaction throughput
- ▶ Industry-leading security and quantum-safe encryption
- ▶ The ability to process 70% of the world's transactions by value, including financial services, reservation systems, and insurance

With a long history of innovation, the IBM Z platform continues to evolve and modernize to meet and often exceed today's expectations for processing speed, transaction throughput, security, and reliability. The recent availability of IBM z17™² marks a major change. It is the first mainframe that is fully engineered for artificial intelligence (AI). IBM z17 provides the processing capacity to perform AI inference on transactions at speed and scale, enabling more than 450 billion inferencing operations per day³ (a 50% increase over IBM z16®) with response times as low as 1 millisecond.

¹ Source: <https://www.ibm.com/z/resiliency>

² For more information, see the IBM z17 announcement at <https://www.ibm.com/docs/en/announcements/z17-makes-more-possible>.

³ Sources: <https://www.ibm.com/z/artificial-intelligence> and <https://www.ibm.com/z/artificial-intelligence#footnotes1>

1.1 Driving forces behind modernizing the IBM Z platform

Several business factors are driving the modernization of the IBM Z platform:

- ▶ **Expanding IT infrastructure:** The IT infrastructure surrounding the mainframe, also known as *systems of engagement*, is rapidly growing. Modern, open application programming interfaces (APIs) enable hundreds of thousands of external applications to interact with the mainframe, which is also referred to as *systems of record*.
- ▶ **Increased security risks and regulations:** The growing engagement with the mainframe platform introduces new risks, such as data leaks and cyberattacks. To mitigate these risks, regulations and elevated security standards for encryption and outage prevention are being implemented.
- ▶ **Growing millions of instructions per second (MIPS) consumption:** As more data is processed on IBM Z, MIPS consumption on the platform has steadily increased over the past few decades.
- ▶ **Adoption of cloud-native concepts:** As businesses require greater agility, you see increased adoption of cloud-native concepts and standards, such as containers and OpenTelemetry, and tools like modern continuous integration and continuous deployment (CI/CD) pipelines on IBM Z. This shift influences how applications are developed, deployed, and operated on the platform.
- ▶ **Shift in human resources investments:** Reduced investments by human resources to hire college graduates and upskill people in favor of distributed platforms have led to a decline in mainframe experience in application development, administrative, and operational roles.
- ▶ **Growing skill gap due to demographics:** The IBM Z platform faces a growing skill gap due to demographic changes because experienced staff in application development, administrative, and operational roles have retired or will retire in the coming years. Hiring and educating people on mainframe skills is not keeping pace with this trend.

These factors affect how the mainframe is operated. As a result, operations, including the tools, processes, and personnel that are responsible for them, must adapt to these changes. Over time, operational silos have been established, each using its own tools and best practices to optimize within its span of control. However, while companies are rapidly adopting cloud-native approaches in application development, the adoption of similar practices in operations has been slower.

Challenges

Enterprise IT infrastructure has evolved over the past three decades. A traditional data center has become an interconnected mesh of platforms, including private and public clouds, platform-as-a-service (PaaS), software-as-a-service (SaaS), and edge compute and storage.

Figure 2-1 illustrates this complexity.

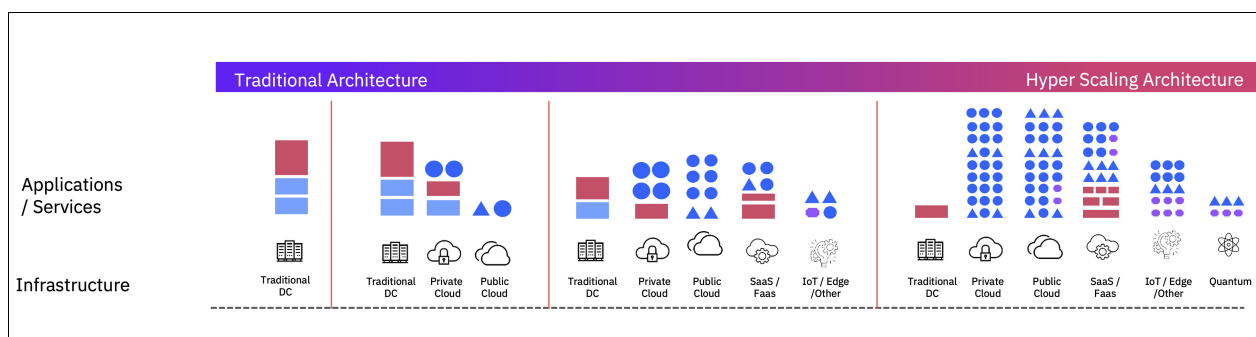


Figure 2-1 Infrastructure growth

As infrastructure grows, the effort that is required to manage and operate it increases. At the same time, IT organizations face constant pressure to balance performance, cost, and flexibility, which adds to the complexity.

Figure 2-2 illustrates the correlation between manageability and complexity.

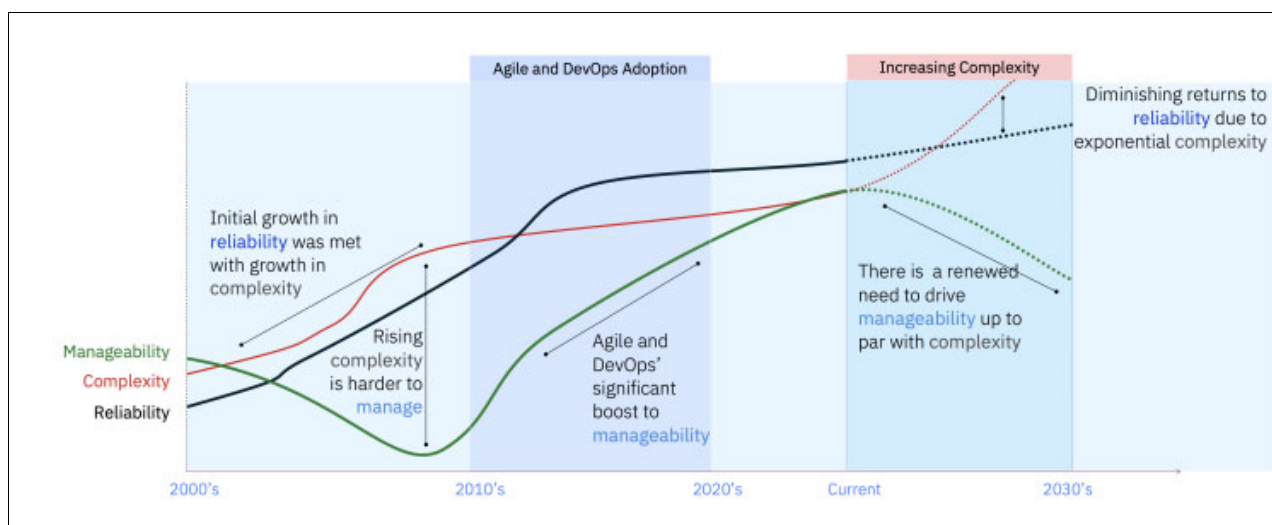


Figure 2-2 Correlation of manageability and complexity of infrastructure

The heterogeneity of platforms and hybrid business applications creates challenges for teams that are responsible for seamless operations. Multi-tiered applications are not new; they have existed since the introduction of client/server architecture in the early 1990s. Organizations have faced heterogeneity since then.

However, the way IT services are used changed over the past 15 years. Cloud-based services are not always less expensive, but they offer greater flexibility to lines of business. These services enable lines of business to use and pay only for what they need, when they need it, without relying on a central IT department.

In contrast, the mainframe has traditionally been managed by the central IT department, which often is perceived not as an internal service provider but as a cost center. This perception triggered a shift toward growing IT services outside the mainframe and increasing budget pressure within IT departments.

The following sections describe some of the challenges that mainframe customers face.

2.1 Too much data and siloed organizations

Managing today's IT landscape, especially the mainframe, is not limited by a lack of data. The mainframe's integrated system instrumentation provides extensive management data for nearly every aspect of operations. However, this abundance often becomes noise. Operations teams face a flood of SYSLOG messages, threshold-triggered exceptions, abnormal metrics, and other sensor data. With limited insight and context, they struggle to correlate events, prioritize effectively, and focus on the ones that are most critical to meet service-level agreements. As a result, the likelihood increases that an important event is detected late or not at all, which negatively affects time to resolution.

This issue is amplified by the siloed structures that persist in many IT organizations. Often, a literal wall separates distributed and mainframe teams. Even mainframe teams are siloed to some degree. Subsystems such as Information Management System (IMS) or IBM Customer Information Control System (IBM CICS®) form their own domains and require expert knowledge to administer, optimize, and scale them to meet the requirements of the business applications running on them.

Over time, a tool ecosystem evolved within each of these domains, contributing to the growing volume of data. Navigation across these domains is further hindered because the tools are often inconsistent in their user experience and require expert knowledge to use effectively.

2.2 Lack of observability

Observability is the ability to understand the internal state of a complex system through its external outputs, specifically telemetry. It enables automated, real-time problem identification and resolution across modern hybrid applications. As hybrid IT environments grow in complexity, observability becomes a critical business capability for helping ensure reliability and manageability.

Enterprise observability, including IBM Z, is now a priority for IT executives. The IBM [Mainframes as Mainstays of Digital Transformation](#) report shows that 91% of IT leaders support hybrid architectures with workloads spanning cloud, mainframe, and edge. However, according to the Kyndryl [2024 State of Mainframe Modernization Survey](#), although 92% of respondents value a unified dashboard for hybrid operations, 85% struggle to achieve it.

Design research among mainframe users confirms the need for seamless integration of IBM Z into enterprise observability solutions. Without this integration, organizations face challenges in troubleshooting, optimizing performance, and proactively detecting issues across hybrid applications, especially ones running on IBM z/OS.

Hybrid applications, such as mobile banking applications, are complex and consist of hundreds of services that are deployed across various landing zones. A mobile application might run on a container service on a public cloud, use Financial Technology (fintech) services, rely on application services that are offered on a private cloud, and depend on transaction processing on the mainframe.

Stakeholders, such as application owners, site reliability engineers (SREs), and application developers, require insight into processing on z/OS to observe the application end-to-end and help ensure smooth operation. Without effective observability that includes the mainframe as a part, they cannot quickly determine whether z/OS middleware services cause or contribute to a particular issue and might perceive the mainframe as a black box. When an incident occurs, multiple teams are often asked to join a conference call or enter a control room to identify the failing application services or infrastructure components. Each team uses its own toolset, which often does not integrate with other toolsets. Many organizations have multiple observability and monitoring tools, each focused on a particular domain of the application, but none provide an integrated end-to-end perspective. This situation leads to slow incident response and increases Mean Time to Recovery (MTTR). It can also create a culture that is focused on proving that an incident is not caused within one's own area of responsibility, often referred to as Mean Time to Innocence (MTTI). Such a culture is counterproductive because it prevents collaboration on resolving incidents as a team.

Mainframe subject matter experts (SMEs) benefit from observability in similar ways to other stakeholders. They can understand which applications generate requests to the mainframe, respond more effectively to changing volumes in application requests, and provide guidance to application teams about how to use mainframe services more efficiently. With observability and end-to-end visibility that include the mainframe, organizations can respond proactively to issues and benefit from advanced techniques such as anomaly detection, automated root cause analysis, and incident remediation.

2.3 How to optimize for availability and performance

IBM Z servers are designed for exceptional availability through extensive hardware and software redundancy. Each system includes spare processor cores that can be dynamically activated if there is a failure to minimize disruption. Memory subsystems incorporate redundant array of independent memory (RAIM) technology, which safeguards large volumes of in-use data against memory chip failures.

IBM Parallel Sysplex® enables high availability (HA) clustering by enabling multiple systems to share data in real time. This architecture helps ensure uninterrupted workloads even if an entire system becomes unavailable. Participating systems can be geographically dispersed to provide extra resilience and flexibility in enterprise deployments.

This level of geographic flexibility and fault tolerance forms the foundation for deeper performance analysis, which focuses on two key areas:

- ▶ Optimizing system and application performance
- ▶ Optimizing resource consumption

At times, these two goals might appear contradictory. Many performance issues are resolved by providing more resources, a common approach in cloud and commodity server environments. Although IBM Z can dynamically increase capacity through functions such as Capacity on Demand, these actions should include management approval. The scalability and proportional value of mainframe components are significant compared to commodity hardware. Using a million-dollar solution for a minor problem is inefficient.

Resource costs share similar proportions. Although distributed software might be billed by the server, IBM Z software is typically billed by consumption, either millions of instructions per second (MIPS) or million service units (MSU). Therefore, resources should not be distributed frivolously.

Availability often correlates directly with performance. For example, if a client taps a card to make a purchase, the system is expected to respond within a few seconds. If the response is delayed excessively, such as for minutes, the application is effectively unavailable, even if it is technically up.

IBM Z is renowned for its unmatched performance and availability. It scales unlike any other platform and can tolerate years of neglect while continuing to deliver. This resilience might create the illusion that systems are running at peak efficiency. Even the most scalable, available platform benefits from professional care and maintenance. Today, many of these professionals are approaching retirement. Mainframe data centers should continue to invest in hiring and training the next generation.

2.4 Increased focus on compliance and vulnerabilities

Regulations such as the [Digital Operational Resiliency Act \(DORA\)](#) in the European Union and the [Business Continuity Management](#) booklet from the Federal Financial Institutions Examination Council (FFIEC) IT Examination Handbook create more challenges for IT providers, especially ones that deliver critical IT services.

These regulations require IT providers to establish comprehensive risk management processes and address vulnerabilities that are identified during major incidents. In addition, providers must define service continuity guidelines that describe how they plan to sustain critical operations, such as incident management, during an outage.

These requirements also affect operations teams, which must adapt to and apply the organization's established guidelines in the following areas:

- ▶ Incident management
- ▶ Change management
- ▶ Automated operations (AO)
- ▶ Backup and restore
- ▶ Disaster recovery (DR) and regular tests

2.5 Cost and speed

Nearly all respondents' organizations (89%) are modernizing their mainframe applications, and 62% reported that these efforts are critical or important to achieving business goals¹.

According to the Kyndryl [2024 State of Mainframe Modernization Survey](#), 53% of respondents' organizations increased mainframe usage in 2024, and 49% expect their mainframe usage to increase in 2025.

Mainframe data centers are often perceived as cost centers. There is constant pressure to reduce spending on personnel, hardware, and software. What do these reports indicate? Although the reports are not directly related, it is reasonable to assume that business objectives include reducing costs or keeping costs flat while workloads grow. Organizations can achieve this goal by investing in application modernization and adopting a modern DevOps approach with continuous integration and continuous deployment (CI/CD) pipelines.

Modern application development increases the ability to develop applications faster and release updates more frequently to production. This approach aligns with the DevOps principle of shifting left to reduce overall development cycles and deliver new business innovations faster.

However, mainframe IT departments are often highly regulated. Their goal is stability to protect against outages and help ensure continuous service delivery. New releases are not rolled out directly into production. Each organization must define and follow a process for how development hands over applications to IT Operations (ITOps) to help ensure a smooth transition into production.

¹ Source:
https://modernsystems.oneadvanced.com/globalassets/modern-systems-assets/resources/reports/advanced_modernization_report_4th-annual-mainframe-modernization-business-barometer-report.pdf

There is no data to show exactly where investments are flowing. However, it is important to emphasize the need to invest in application modernization, ITOps, personnel skills, and modern management tools. Regardless of how robust a mainframe is or how much redundancy is in place, failures can still occur due to human error, oversight, or software defects, and these failures can suddenly impact many business-critical workloads. A trained operations team that understands the infrastructure and is skilled in the operating system and subsystems is critical to restoring service as quickly as possible. The savings are significant; according to the Information Technology Intelligence Consulting (ITIC) [2024 Hourly Cost of Downtime Report](#), enterprises pay between \$1 million and over \$5 million per hour of downtime.

The lack of experienced professionals and limited investment in their education directly affect the last challenge: the mainframe skill gap.

2.6 Mainframe skill gap challenges

As new talent joins mainframe departments, the typical journey that they experience is shown in Figure 2-3.



Figure 2-3 Early tenure journey

According to the Kyndryl [2024 State of Mainframe Modernization Survey](#), 53% of new hires entering the workforce have no mainframe skills. As a result, junior operators or SMEs often need extra time to find the right information for their tasks. In addition, with 49% of experienced staff retiring, the ability to find experts who can interpret and resolve problems decreases.

Many companies (86%) report investing in the deployment of generative artificial intelligence (AI) tools and applications on the mainframe. However, the growing interest in AI should not overshadow the fact that human resources departments have neglected to hire mainframe staff and invest in their education for many years.



Artificial Intelligence for IT Operations capabilities

Everyone talks about Artificial Intelligence for IT Operations (AIOps). But what does it mean? A key part of the name is the word *Ops* or *Operations*, which is an abbreviation of the more precise term *IT Operations* (ITOps). Running an IT business means operating the IT infrastructure so that the applications that are deployed on the infrastructure run at the service levels that are agreed to by the IT service provider and the IT service consumer. This statement is generic and applies across all types of IT infrastructure. Whether cloud applications, classic mainframe applications, or distributed hybrid cloud applications, the service that an application provides is measured against service-level objectives. Operations are responsible for helping ensure that the service-level objectives are met. Operations help ensure that the required compute, storage, and network resources are available. Operations must detect when actual service levels are at risk or not being met. They must quickly identify and resolve issues to restore service levels as soon as possible.

The second part of the name is artificial intelligence (AI). Asking different stakeholders often results in different answers. This IBM Redbooks publication uses this definition:

AI refers to the capability of computational systems to perform tasks that are typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

In the context of operations, AI can reduce the complexity of the ever-growing IT landscape. It can automatically analyze the thousands of signals that applications or the underlying IT infrastructure generate and uncover anomalies. It can help prioritize events to reduce noise. AI can also provide background information and remediation tips when operators must understand an issue and how to fix it. Over time, AI can learn the specific behaviors of applications and IT infrastructure, based on seasonal and human behaviors to provide more accurate predictions and remediation recommendations. Embedded in tools, AI can reduce human effort for manual instrumentation, such as setting thresholds, and can help detect impending resource bottlenecks or other issues early to prevent outages.

With these challenges present, a structure is needed to help clients and software vendors identify the essential capabilities for operations and map existing or wanted products to those capabilities. IBM established AIOps on the IBM Z framework around three pillars to structure its portfolio in the ITOps management space, which are shown in Figure 3-1:

1. Detect: The ability to have visibility into all IT services and their underlying infrastructures to identify issues or anomalies at the earliest possible point in time.
2. Decide: The ability to analyze issues and anomalies by drilling into details and bridging into different management domains (for example, from z/OS to Db2) until the root cause is found. It also includes the ability to classify the vast amount of data so that analysts can plan for future resource demand and optimize existing resources.
3. Act: The ability to resolve issues rapidly, reliably, and with confidence to reduce business impact.

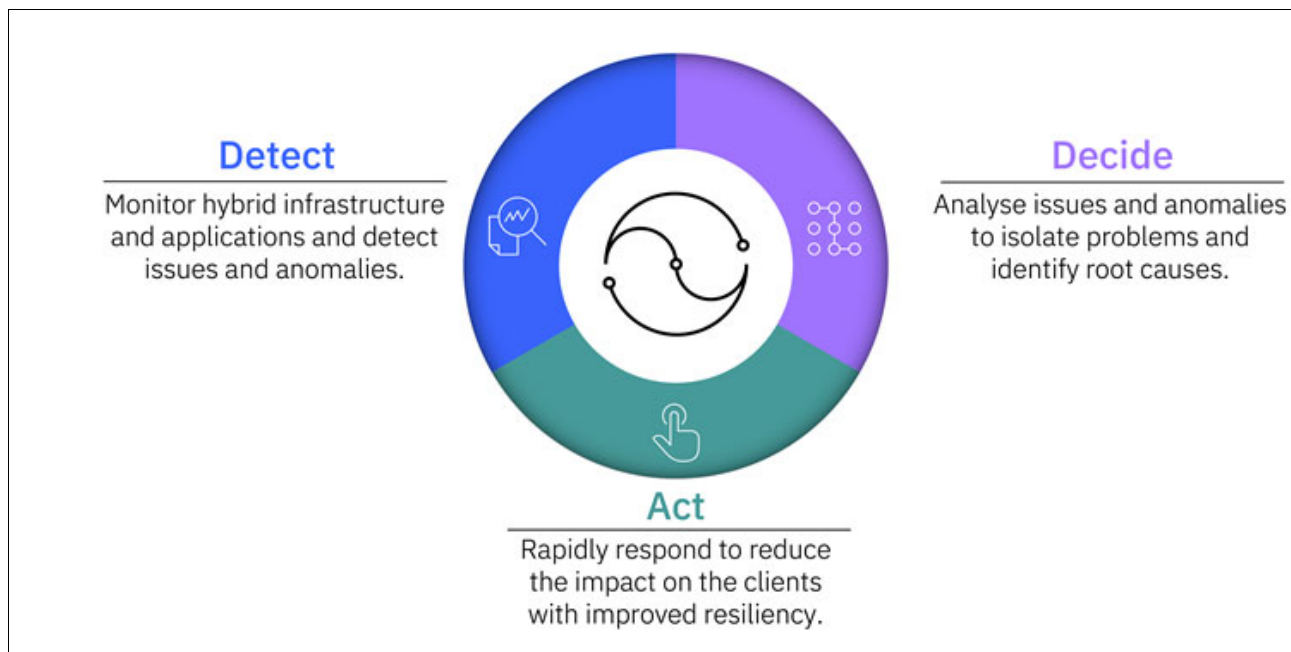


Figure 3-1 The AIOps on IBM Z framework

These capabilities do not work in isolation. The AIOps on IBM Z framework tightly integrates them through dashboards and collaboration tools to enable collaborative incident management. The goal is to reduce the time to detect, isolate, and remediate incidents. This integration connects the different disciplines, bridges gaps that are caused by organizational isolation, and ultimately determines how effective operations can be.

The following sections and the remaining chapters illustrate the most important capabilities that evolved over more than 60 years of IBM Z history and its predecessors. Core capabilities that are required for the use cases are listed, along with solutions that address the challenges outlined earlier.

3.1 Industry background: IT service management and ITOps

This section provides background about how the industry (including analysts, IT providers, and clients) defines operations and its role within overall IT service management (ITSM).

3.1.1 IT service management

In essence, ITSM is the practice of planning, implementing, managing, and optimizing the delivery of IT services to meet user needs and business goals.

ITSM relies on software tools, automation, and procedures. For example, if a branch office contacts the central IT department to report a problem with the reservation service, ITSM defines the process and guides the workflow to resolve the problem. In this example, the situation is formally referred to as an incident, and the process is referred to as incident management.

Various ITSM software solutions and guiding frameworks exist for implementing ITSM. ITIL (formerly known as Information Technology Infrastructure Library) is one of the most widely used frameworks in the industry. It defines processes and functions that are deployed throughout the lifecycle of a service, from strategy through design to moving into daily operations, and finally to the operation of the service itself. ITIL provides an over-process for continuously improving services to enhance user experience and business productivity.

3.1.2 ITOps

While ITSM describes the various IT services and how they interact, ITOps focuses on fulfilling these services. Following ITIL, ITOps is a function under the Service Operations phase in the service lifecycle. It includes all activities from implementing, managing, delivering, and supporting IT services to meet business needs.

Customers and businesses expect applications and IT services to be immediately accessible. However, this expectation is only possible if these applications and services, including the underlying hardware, software, and network infrastructure, run at optimal performance. Even short IT outages can impact business operations and become costly. Therefore, the primary role of ITOps is to help ensure that all IT resources perform as expected so that business operations proceed without interruption.

Here are the main responsibilities of ITOps:

- ▶ **Managing resources**
ITOps keeps IT infrastructure (hardware, software, and networking) and applications running. It provisions and manages IT infrastructure and maintains service delivery to application teams, customers, and partners.
- ▶ **Optimizing IT infrastructure**
ITOps strives for effective resource usage while reducing cost safely. This responsibility includes planning and documenting configurations, workloads, hardware and software upgrades, and the impact of configuration changes.
- ▶ **Helping ensure application performance**
ITOps collaborates with line-of-business and application owners to help ensure that sufficient resources are assigned to applications to prevent slowdowns or outages.

- Service desk support

ITOps often provides IT support, which includes managing the help desk and ticketing system, troubleshooting issues, and addressing the root cause of IT-related problems.

- Incident and security management

Although the focus is on day-to-day availability of IT services, ITOps performs regular data backups, restores systems after maintenance or outages, and provides plans to safeguard future availability of systems if problems arise.

3.1.3 Incident management

This section provides an example of ITOps by looking closer at incident management. In general, an incident management process uses the capabilities that are shown in Figure 3-2.

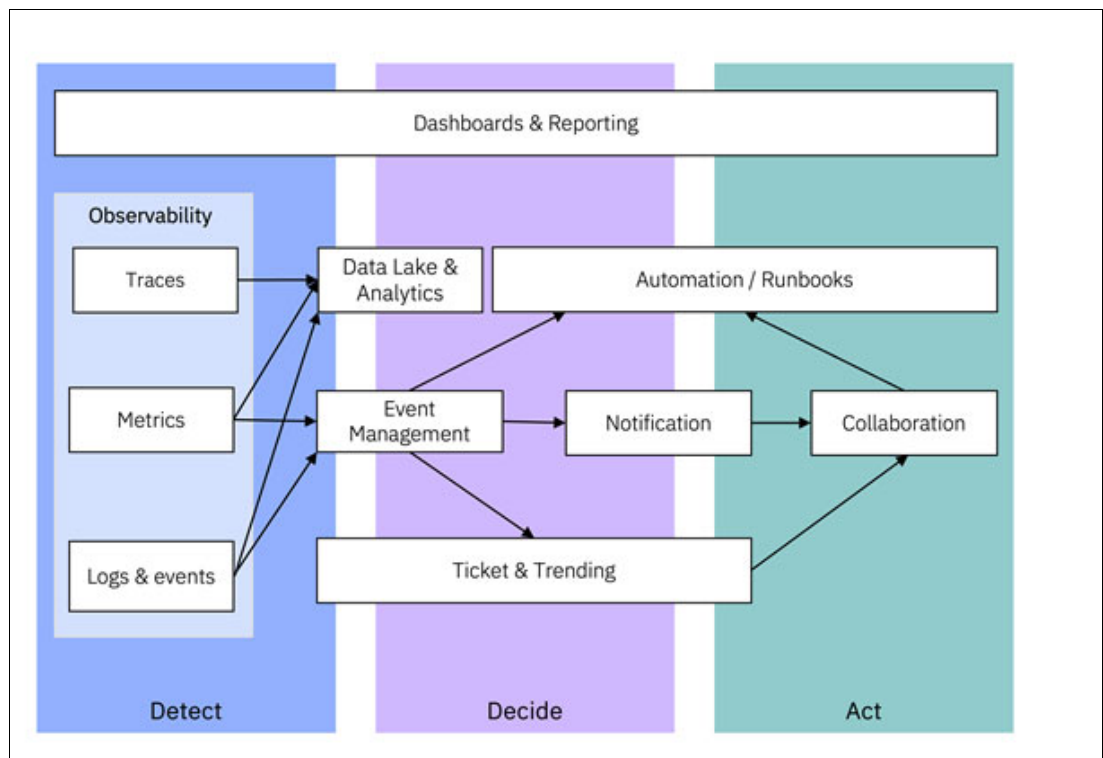


Figure 3-2 Incident management

Within operations, incidents are considered as a response to an event. According to ITIL, an event is a detectable and discernible occurrence that is relevant to IT service delivery and indicates the impact of degradations or disruptions.

Events that represent an abnormal observation might not lead to an outage of a function, component, or application, but they can be the precursor to more severe damage if ignored. Therefore, it is important to detect such events as early as possible and respond by creating an incident to analyze the occurrence and resolve it.

Clients typically follow these general phases to resolve incidents:

1. Detect: Observe what happens.
2. Decide: Analyze the cause and plan remediation.
3. Act: Run the remediation plan.

Detect

Operators and subject matter experts (SMEs) need specific tools throughout these phases to work effectively on an incident. The insights from one tool are applied by the next. For example, a key performance indicator (KPI), such as MVS CPU usage, collected periodically, is added to the data lake for later time-series analysis. If the same KPI exceeds a user-defined threshold, it generates an event. An event typically marks the starting point for working on the incident because it is when you first become aware of an issue. As a result, the system informs an operator and instructs them to begin the analysis.

Events can originate from metrics, logs, and traces.

► Metrics

The mainframe is an instrumented compute platform. No other platform generates measurement data as broad and detailed as z/OS. The operating system provides thousands of metrics that are stored in data records that are created by the System Management Facility (SMF) at specific points in time or periodically at the expiration of a customer-defined interval. Middleware subsystems and certain management applications provide more metrics and information representing the workloads that they manage. These SMF records form the foundation for performance analysis and capacity planning because they provide insights about z/OS and its middleware subsystems (IBM Customer Information Control System (CICS), Db2, Information Management System (IMS), and IBM MQ) that process batch and online transactions.

In addition to SMF records, online monitors provide real-time or near real-time insights into various aspects of the system through an interactive user interface. Like the instruments in an airplane cockpit, they show issues that require immediate attention and through their interactive nature, enable SMEs to perform root cause analysis and remediate the issue immediately.

► Logs

z/OS and all workloads running on it produce large volumes of log data that are essential for auditing and problem determination. Modern applications, including ones on z/OS, often use a logger library that is part of the programming language ecosystem (for example, log4j for Java). Native z/OS applications write their output into the z/OS Job Entry Subsystem (JES) job log or sometimes as Write-To-Operator (WTO) messages in the z/OS SYSLOG.

► Traces

Traces help operations teams understand how software components interact, particularly how transactions flow through the participating application tiers. Static relationships and dynamic traces can be distinguished. Sometimes, an analyst only needs to understand how different middleware components relate to each other. This information can be discovered by inspecting the active address spaces on z/OS. By using characteristics of these components, it is possible to infer which components communicate with each other.

When analyzing application bottlenecks, especially end-to-end, dynamic tracing is necessary. An observability tool is required to generate trace spans at defined points along a transaction's path through the system. These trace spans can be collected and sent to an observability back end to visualize the overall application request flow from its origin, across distributed platforms, into z/OS, and within z/OS. Distributed tracing is covered in more detail in a later section.

When the data lake is on a different system than where the data is produced, you need a tool that can optionally filter and transform the data into a standard format, such as OpenTelemetry Protocol (OTLP), and relay it to its final destinations. Filtering the data reduces network traffic and costs by minimizing the footprint of data at rest.

Decide

In this phase, the operator or SME tries to determine the root cause of an event. The most important capability in the Decide phase is the analytics platform and its associated data lake, where all observable data (metrics, traces, and logs) is stored.

During the analysis, event management plays a key role. According to ITIL, event management is one of the processes in service operations, which is one of the five phases of ITIL service management. Based on experience, every mainframe client follows an event management process and uses tools that help detect, filter, classify, and correlate events. Events that require further analysis are addressed by creating an incident (or ticket) so that the analysis steps and insights can be documented and archived for auditing and future use. Event management is often handled at the enterprise level, where events from the entire IT landscape are centrally managed.

Tools for creating, updating, and harvesting historical incident data are essential for effective incident management. Many IBM mainframe clients use a third-party ITSM tool such as ServiceNow. An ITSM tool serves as a global repository for service management data. The core of that repository is the Configuration Management Database (CMDB), where all configuration items (hardware, software, network, and services) that are relevant to IT are maintained. Other components include repositories for incidents, problems, and changes. A seamless incident management process requires that all software tools that are used to facilitate the process use and update these repositories to automatically identify the impacted configuration items and business services.

Based on existing input and insights from the analysis, the necessary action must be determined. The incident and its related ticket provide context about the original event and what occurred so far during the analysis. If the root cause is known, the incident can be resolved by identifying and running the necessary remediation steps. If the incident occurred before, the operations team might already have a runbook (automated recipe, script, and other process) for resolution. If the root cause is not yet known, the next action might be to involve an SME for further investigation.

Involving other specialists, such as a CICS expert to analyze a CICS region's storage consumption, requires notification. The method of notification depends on the ITSM tools in the enterprise. Mainframe teams rarely determine these tools, which makes it important that tools running on z/OS can integrate seamlessly with enterprise tools.

Here are some examples of how to notify operations teams and SMEs:

- ▶ A messaging service such as PagerDuty, a common third-party provider
- ▶ Proprietary webhooks into the enterprise event management tool
- ▶ Notification through Event Integration Facility (EIF) events, an older but still widely used mechanism
- ▶ Email

Act

In the Act phase, the operator performs the necessary activities to remediate or resolve the issue. Depending on the nature of the incident, these activities might consist of written step-by-step instructions that run manually, predefined automation, or a combination of both.

Chat operations (ChatOps) facilitates collaboration and accelerates software development and incident resolution. Many mainframe users struggle with effective cross-team or remote collaboration and often find it difficult to locate missing information or the right expert. Using a chat-based collaboration tool, already familiar in day-to-day work, and extending its use to operations addresses these challenges.

Collaboration tools such as Slack, Mattermost, and Microsoft Teams provide a communication platform for exchanging messages between individuals, groups, or channels. One of their strengths is the ability to configure bots that can be triggered by users to perform tasks. In operations, such tasks might include querying jobs waiting for running or recycling a CICS region to apply new configuration parameters. ChatOps enables collaboration among different people and experts in mainframe operations, and enables automation of certain tasks without requiring deep expertise. The bot offers a chat interface to initiate a dialog with the user, respond with targeted answers, and provide options for continuing the conversation.

Companies usually maintain a catalog of activities describing how to resolve incidents based on prior experience. Automation that is designed for operators to repair recurring issues is often referred to as a *runbook*. An Ansible playbook that increases the size of a data set in response to a B37 system abend is an example of such a runbook. Operators can find the playbook in the corporate operations handbook for handling abends that are caused by exhausted data set extents. An AI assistant that is queried in chat can reference the operations handbook and even offer to run the playbook immediately. Another example is a predefined script or command that starts through the automated operations (AO) product to resolve an unexpected JES SPOOL¹ shortage. Yet another example is triggering an ad hoc batch application through the job scheduler to re-create a corrupted database index.

Summary: Closing the loop

Figure 3-2 on page 12 shows an iterative flow in practice. Operators use different tools to identify the root cause. They examine related metrics and z/OS log data and collaborate with various SMEs depending on the scope of the incident. Middleware experts contribute their tools and insights to trace transaction flows. Together, they continue investigating the issue until the root cause is identified. Throughout the process, the incident is represented by the incident ticket, which tracks what is impacted, how it is impacted, what investigations were performed, and what actions were attempted with their results. When the incident is resolved successfully, the associated ticket is closed and archived for auditing and future research.

Across the phases, operators and SMEs use dashboards that facilitate analysis and help users understand the reach of an event, sometimes referred to as the *blast radius*. Dashboards are designed to guide users and present information in a way to find relevant details based on their role. They support context-sensitive drill-down into details, include topology visualizations, and enable users to act on the entities that are depicted.

The emerging use of ChatOps² further benefits operations because users can use their enterprise chat-based collaboration platform (for example, Slack or Microsoft Teams) to collaborate with colleagues and interact with tools through application programming interfaces (APIs). Augmented with AI capabilities, ChatOps enables conversational AI, enabling users to converse in natural language with built-in assistants, answer questions, perform automated analysis, and receive recommendations. In environments such as mainframe operations, where organizational silos developed over many years, ChatOps can help with information access and exchange among operations staff, contributing to faster incident resolution.

¹ Simultaneous peripheral operations online, which is a set of datasets for jobs and job output.

² For collaboration benefits, see <https://www.ibm.com/think/insights/benefits-of-chatops>.

3.2 Observability and monitoring

This section describes the basic capabilities of all ITOps. The terms *monitoring* and *observability* are defined, along with their importance for successfully operating a mainframe. Both monitoring and observability are required capabilities to follow the incident management process that is described in 3.1.3, “Incident management” on page 12.

Observability can help streamline incident management because it detects events. It also facilitates understanding of the data and gaining insights that can be used to take corrective actions.

3.2.1 Defining observability

Observability is the ability to gain actionable insight into complex systems. It enables users to understand the state of a system from external outputs and take corrective actions. Observability enables detection of issues and helps identify their causes to proactively resolve them before they impact customers. It is built on the correlation and analysis of several types of signals: monitoring metrics, logs, distributed traces, structured events, relationships and dependencies, contextual metadata, and resource consumption profiles.

Observability enables users to answer questions about the customer experience and the underlying components that contribute to an issue:

- ▶ What did the customer experience?
- ▶ Which customers are impacted by the issue?
- ▶ What is the impact on the business and revenue?
- ▶ What other systems, processes, or configurations contributed to that experience?

Modern observability platforms increasingly rely on AI-powered capabilities for anomaly detection, root cause analysis, and predictive insights to enable proactive issue resolution through automated remediation actions. Design research among mainframe users highlights the need to integrate the mainframe into enterprise observability. Use cases based on real-time transaction processing, such as instant payment applications, drive the requirement for end-to-end observability because they demand immediate insight into application behavior and the underlying infrastructure that supports it.

The distribution of mainframe telemetry data must be simplified. New applications and regulations require mainframe telemetry data to be more accessible, usable, and shareable across the enterprise. Organizations are adopting OpenTelemetry, an open standard for observability, as the unifying telemetry data layer of their enterprise observability architecture. These organizations request OpenTelemetry support on the mainframe to unify telemetry distribution across all landing zones of their IT environment and benefit from flexibility in choosing observability tools.

3.2.2 Defining monitoring

Monitoring is the process of collecting and analyzing predefined metric data from systems to help ensure their health, performance, and availability, and to act on the data to prevent and resolve customer-impacting issues. Monitoring enables users to answer questions about system performance, resource usage, system uptime, batch job completion, and threshold compliance.

Monitoring is a capability of observability. Observability adds context and correlation with other signal types to the metric data, enabling users to ask questions about data that was not initially monitored. Observability depends on monitoring to track system activity, identify problems, and determine the root causes of issues. z/OS offers exceptional built-in instrumentation, facilitating reliable monitoring with operational maturity across various levels of granularity, from minute intervals to real time.

Monitoring provides an inside-out perspective, and observability offers an outside-in view. High CPU usage that is reported by a monitor does not by itself and without context indicate a problem. In fact, MVS runs efficiently even at close to 100% usage. However, with observability, an alert about degraded performance of instant payments suggests that high CPU usage might be the cause and that more capacity is needed.

The operating system supports real-time monitoring, metric aggregation at configurable intervals, and reporting of historical monitoring data. SMF is the primary component that powers z/OS monitoring capabilities. SMF enables data collection and recording and serves as a source of performance data, accounting, workload management, and capacity planning. Modern monitoring solutions ingest and analyze SMF data alongside other sources. For users in various roles, monitoring remains a primary method to determine whether the system is operating within expected parameters. Advanced monitoring techniques enable users to identify system situations statically. When anomalies are detected or thresholds are breached, alerts are triggered. The monitoring system can act to correct the situation or initiate resource adjustments to maintain system performance and availability.

Historically, z/OS monitoring began in the early days of mainframe computing with batch reports. These reports were created from system data as formatted text and tables and run periodically in batch jobs. Over time, z/OS monitoring evolved into multiple categories to meet the needs of different users. For example, z/OS system programmers rely on performance metrics for CPU, memory, I/O, and workloads to manage LPARs and system resources efficiently and maintain z/OS subsystem health. Monitoring workload management metrics such as performance index, response time, velocity, and processor delay enables system programmers and capacity planners to assess how well service goals are met, identify performance bottlenecks, and prioritize important workloads for efficient operation. Job monitoring enables operators to oversee job execution, queues, and batch throughput of the z/OS JES subsystem. Transaction and subsystem monitoring provides detailed metrics about resource usage during transaction processing, including queues, tasks, and databases. Detailed metrics are available for each z/OS subsystem to analyze its behavior and transaction processing steps within its scope. z/OS monitoring collects metrics on every aspect across an IBM Parallel Sysplex, including workloads, member systems, and underlying infrastructure. The metric data is accessible through interfaces such as REST APIs and telemetry pipelines and can be visualized through interactive dashboards for an operational view of system health, performance, availability, and resource usage.

3.2.3 Establishing visibility into z/OS

To support modern AIOps practices and enterprise observability strategies, visibility into z/OS must evolve beyond traditional monitoring. This section introduces the foundational elements that are required to achieve comprehensive visibility into z/OS and enable seamless integration with enterprise observability platforms. These elements include capabilities such as distributed tracing, telemetry correlation, and unified telemetry pipelines, which contribute to a holistic understanding of system behavior. The following subsections explain how these components work together to provide actionable insights and facilitate end-to-end observability across hybrid environments.

End-to-end observability

Observability relies on telemetry from the signal types that are described in 3.1.3, “Incident management” on page 12 during the Detect phase of incident management to provide a comprehensive understanding of system health. End-to-end observability focuses on enabling visibility into all services of a hybrid application. When application services are deployed across multiple landing zones, such as public cloud and a mainframe environment, users of enterprise-wide observability solutions require as much insight across all services and landing zones for an efficient incident response. Enterprises with an application-centric approach to observability aim for a unified observability platform that offers users a single-pane-of-glass view across the complete application landscape. IBM Z and z/OS integrate into such enterprise observability solutions as first-class participants.

z/OS provides rich data sources for different signal types. Enabling end-to-end observability for z/OS involves collecting, aggregating, and distributing telemetry data from these sources to an observability back end. A holistic approach goes beyond traditional monitoring and integrates distributed tracing, enriches telemetry with context, and correlates diverse signals for a comprehensive system view. For example, by correlating latency data from the spans of a distributed trace with infrastructure metrics of the corresponding z/OS resources, a site reliability engineer (SRE) can determine whether an increase in request latency is caused by an overused z/OS component.

When defining their enterprise observability strategy, many organizations adopt open standards such as OpenTelemetry. The OpenTelemetry project aims to make high-quality telemetry available for effective observability and unify the collection, processing, and distribution of data from any source to any target. Organizations have different approaches for adopting OpenTelemetry in their enterprise observability strategy, and some rely on OpenTelemetry as a unified telemetry data layer across the hybrid cloud environment. For z/OS, various integrations exist with different proprietary observability back ends. OpenTelemetry enables mainframe clients to unify telemetry distribution from z/OS and reduce dependency on proprietary protocols.

Distributed tracing for z/OS

Distributed tracing records the journey of user requests through a hybrid application. Traces show the services that are involved in processing user requests and provide information about the latency and dependencies of each service. The services participating in request processing are represented as spans within the traces. Distributed tracing enables dynamic discovery of the relationships between application services and displays them in tools such as service graphs or maps. Traces are crucial for understanding performance bottlenecks and issues in complex hybrid applications.

Tracing requests across multiple services, including z/OS system software such as CICS or IMS, enable SREs to understand transaction flows and gain insights into processing on z/OS. The resource information and attributes that are associated with spans of the z/OS services enable the SRE to hand off an incident to a z/OS expert, who then has sufficient information if an incident analysis requires deeper investigation of a z/OS subsystem. Distributed traces have a different focus than the traces that are available in z/OS for problem determination. Although z/OS traces provide detailed information about the processing of a program or transaction on z/OS, distributed traces are used to gain end-to-end visibility into the entire hybrid application.

Distributed tracing relies on two features to enable the recording of spans in a trace:

- ▶ There must be a mechanism to detect and correlate the services that are involved in request processing.
- ▶ There must be a method to format and export the spans to an observability back end.

The observability tool is responsible for visualizing the trace with all its spans, the involved services, the hierarchy of calls, and displaying the spans with lengths representing their duration relative to the trace. In addition, users must be able to browse attributes and between traces, spans, and associated telemetry.

The path that a request takes through different services can be followed when extra context is passed from one service to another one. This process is called *context propagation*. Although multiple specifications for propagation exist (some from open-source tools, others from vendor products) the standard that is most widely used today is the [W3C Trace Context](#) specification, which is also the default for OpenTelemetry.

The W3C Trace Context specification describes the HTTP headers and the format that is used to propagate context information. The specification defines the traceparent header, which carries information to uniquely identify an individual request and a parent service that calls a child service. This capability enables service correlation and enables analysts to follow a request end to end.

On z/OS, several types of connections and interfaces are used for communication with external clients and between z/OS system software. Unlike HTTP, headers cannot be directly injected and propagated. Instead, an agent-based approach is used to capture information about incoming requests from z/OS system software. With this information, including the traceparent and other correlators, proper span information is built and exported to an observability back end. More recently, native implementations of distributed tracing are available that support built-in W3C context propagation in z/OS system software.

Telemetry correlation

Correlation of telemetry from different signal types is a key capability for gaining insights into system health. Observability tools typically use one signal type as the primary source. In modern observability platforms, events or traces are often used as the primary signal type, with other types correlated to them.

In an event-centered approach, events are structured, contextually rich records of distinct changes in the system or application state. Events can be correlated with other telemetry data such as metrics, traces, and logs. Structured events are often viewed as the fundamental units of observability because they capture high-cardinality, context-rich data points that facilitate real-time understanding and debugging of systems. In contrast, OpenTelemetry uses traces as the primary signal type and specifies how to correlate traces with other signals. Structured events and telemetry that are gathered by using trace-centered tools complement each other.

Both event-centric and trace-centric methods aim to provide deep insights into system behavior but differ in their entry point for incident identification and analysis and in how users browse telemetry data. From the perspective of a telemetry pipeline, OpenTelemetry instrumentation collects and receives correlated telemetry in the form of traces, metrics, and logs, and structured events are the result of data analysis and processing by an observability back end.

Other approaches use metrics or logs as the primary signal type. These approaches are limited in how telemetry can be correlated. Logs are difficult to correlate across systems unless tied together by instrumentation that also supports traces. Metrics are aggregated measurements over a fixed time interval and might limit visibility into a specific event.

The best correlation between telemetry of different signal types is achieved when instrumentation and the data model explicitly support correlation. In this case, the data model describes how to link telemetry from different signals and provides guidance about setting the identifying attributes for the emitting resources. Other methods rely on temporal correlation or derive correlations based on patterns, such as matching identifiers that are found in telemetry.

On z/OS, identifying attributes for correlation at the resource level are injected by the instrumentation producing the telemetry data. Identifiers to correlate spans of distributed traces with related metrics are also injected at the source. Tools for resource and relationship detection support the process of enriching telemetry at the source. Temporal and pattern-based correlation is often performed in an aggregation layer, where telemetry data is enriched and transformed before being sent to consumers. Observability back ends benefit from correlated signals when deriving insights, querying, and visualizing telemetry data with context.

Unified telemetry aggregation and distribution

Telemetry pipelines play an increasingly important role in enterprise observability. Pipelines enable organizations to specify, in a declarative way, how to send the right data to the appropriate tools and teams. By implementing a unified approach for telemetry pipelines, organizations can enable seamless data flow, help ensure consistency in telemetry data across applications and infrastructure, and streamline processing and distribution to observability back ends and other tools. Pipelines enhance real-time observability capabilities and provide a single point of control over data flows within an organization.

Telemetry pipeline control

Telemetry pipelines support the process of receiving, transforming, and distributing telemetry data. Pipelines describe the path that data takes from a source to a target and the processing steps that are applied along the way. On z/OS, sources can include selected SMF records, the z/OS SYSLOG, or trace spans that are related to z/OS system software. Pipelines support operations based on traces, metrics, and logs. The OpenTelemetry Collector implements support for telemetry pipelines. Typically, a pipeline processes data of one signal type in isolation. However, correlation between signals can be achieved by injecting shared attributes into the signals through processors. Processors provide powerful tools to modify and transform data while it is in flight. Examples of telemetry processing include batching, filtering, sampling, data cleansing, and enrichment. Filtering and transformation steps determine whether z/OS signals and their associated context are exported or skipped. Targets include observability back ends, AI-powered ITOps tools, analytics platforms, telemetry gateways, or any OTLP endpoint.

Several participants are involved in setting up telemetry pipelines for z/OS. Various components on z/OS and the z/OS system software act as telemetry data producers. Producers generate the operational data that forms the telemetry signals (metrics, traces, and logs). Data aggregators perform critical processing steps in managing telemetry data before sending it to consumers. Aggregators handle filtering, enrichment, and other processing of the source data. Tools that act as data aggregators can be on z/OS, Linux on IBM Z, or off-platform. Data consumers use z/OS-generated data for analysis and presentation of insights. These tools include enterprise observability and IBM Z specialized back ends, and security and event management tools.

Telemetry ingestion

Ingestion of z/OS telemetry data into observability back ends and other tools can be done in various ways by using a push or pull model. Clients can query telemetry through REST APIs from monitoring tools or configure them to push data by using OTLP or vendor-specific protocols. Pipelines can be configured to convert data from the source format into the format that is required by the back end. Because OTLP is now established within the observability ecosystem as the open standard for telemetry distribution, configure pipelines that receive z/OS telemetry with an OTLP exporter.

Telemetry transport

Organizations often use Apache Kafka in their observability architecture as a high-performance, fault-tolerant distributed streaming platform to transport telemetry from data producers to data consumers. Integrating pipelines with Kafka enables near real-time ingestion of z/OS telemetry into data consumers. Telemetry pipelines can be combined with Kafka by using an exporter that writes the data to a Kafka topic.

3.3 Performance and capacity management

Applications continue to grow and multiply at an unprecedented rate, often spanning multiple areas of infrastructure within and beyond the IBM Z environment. The demand for new features and functions, when combined with time-to-market priorities, often takes precedence over application efficiency. This challenge is compounded by a unique characteristic of the IBM Z platform: unlike commodity server environments that typically provide dedicated servers or clusters to applications, IBM Z hosts dozens or even hundreds of concurrent applications. Its reliability, scalability, and security make it the preferred choice for mission-critical workloads.

The mainframe is a large, shared ecosystem, and the key word is *shared*. An application's performance might be affected by other applications. Analyzing individual components or data sources might not provide the complete picture. Similarly, common solutions such as increasing priority or adding capacity are not straightforward. Although these actions are possible, they require careful consideration. Increasing one application's priority might affect the performance of others, so relative business importance must always be considered. Adding capacity can resolve many performance concerns, but it is a management decision because these actions can affect software costs.

An IBM Z performance analyst is responsible for balancing the priorities of all applications and managing the provisioning of costly resources. If this decision is left to application owners, each would claim to be the most important, resulting in chaos as they compete for shared resources. To achieve these goals, the IBM Z performance analyst must know a little about everything and a lot about a few areas, including:

- ▶ The machine (for example, IBM z17)
- ▶ The hypervisor (PR/SM)
- ▶ The operating system (z/OS)
- ▶ Core OS components (SYSPLEX, JES, SRM, and Workload Manager (WLM))
- ▶ Core subsystems (CICS, Db2, IMS, IBM MQ, and IBM WebSphere®)
- ▶ The applications themselves

The synchronization of these elements is part of what makes IBM Z the most efficient enterprise computing system in the world. Although there are specialists in each of these areas, the performance analyst must have a working knowledge across all of them.

As stated earlier, the IBM Z platform offers unmatched scalability. It is typically less susceptible to frequent problems than commodity servers. However, when slowdowns or outages occur (though rare), they are often more costly than on other platforms because of the critical nature of applications that are hosted on this premium platform. Proactive analysis and tuning can reduce the likelihood of such costly events.

Beyond direct disruptions, there are two key reasons to focus on performance and capacity:

- ▶ Application performance, which directly affects the client experience
- ▶ Resource consumption, which contributes to software costs

Achieving these goals begins with comprehensive measurements. Only when you can measure something can you improve it.

The IBM Z platform uses instruments. Detailed statistics on every aspect of the platform, from the hardware and operating system to subsystems and workloads, are recorded every second and can be offloaded and stored indefinitely. Historically, a major challenge has been accessing this data. Mainframe data is both vast and complex, traditionally requiring specialized tools and skills to extract and produce reports.

For many years, real-time monitors such as IBM OMEGAMON® provided quick insight into system and application performance. These powerful tools continue to evolve with AI-enhanced anomaly detection and real-time automation. These tactical solutions remain key tools in the analyst's toolkit. Although these technologies offer peace of mind to data center operations, they should not be your only approach. To use an automobile analogy, technological breakthroughs such as anti-lock brakes (ABS) and airbags are critical safeguards to prevent disasters, but that does not mean that you should stop teaching teenagers how to drive.

In a data center, this strategy is supported by long-term tools, such as IBM Z IntelliMagic™ Vision™ for z/OS, which track trends and reveal the effects of task interactions within the IBM Z ecosystem. Tracking and understanding the normal behavior and patterns of your mainframe environment, when combined with institutional knowledge about your business, provides critical insights to develop and continuously improve your business and IT strategies. For example:

- ▶ Is the critical application response time consistent during month-end processing? What about year-end workloads?
- ▶ Are limited resources distributed based on the relative business importance of workloads?
- ▶ Are these resources aligned with consumption-based software bills?

Along with experience, modern AIOps tools provide superior insight into the inner workings of the machine. These tools extract and show the vast and complex data that is recorded by the IBM Z platform and present it in graphical formats. Integration with IBM watsonx® AI helps new users ask questions such as “What is the response time of transaction X?” or “What are the highest consuming tasks in my system?” IBM watsonx provides written responses and presents reports, expanding insight into the IBM Z environment and helping with the skills gap challenge. These solutions reduce complexity and provide modern interfaces, enabling newer staff to quickly understand what is happening inside the modern mainframe.

3.4 Automation

This section explains what automation means from an operations perspective. It describes the types of tasks that are automated and the reasons for automation, such as improving efficiency, reducing manual effort, and increasing reliability. It also provides context on why automation on z/OS might appear different from automation on other platforms. Although the tools and terminology vary, the underlying principles remain the same. Finally, this section examines the specifics of automation on z/OS, with a focus on job scheduling and how it supports continuous, efficient system management.

3.4.1 Overview

Automation is an overloaded term. On the mainframe, *automation* primarily refers to running and monitoring different types of workloads across various operational areas. These workloads include started tasks, batch jobs, UNIX System Services processes, and, starting in 2024, z/OS Containers.

- ▶ One operational area is automation operations (AO), which includes system and application startup and shutdown. This area focuses on automating started tasks and includes processes such as daemons (for example, FTP).
- ▶ Another operational area is workload automation, traditionally known as batch job scheduling. This area helps ensure that business-related batch jobs run at the right time with the necessary resources for successful completion.
- ▶ The third operational area is part of DevOps. It focuses on provisioning and configuring software infrastructure to deploy and run business applications in development, test, or production environments.

Figure 3-3 illustrates the integration of DevOps-driven automation with z/OS-based workload automation and AO. It presents a layered view of how different operational areas interact across platforms.

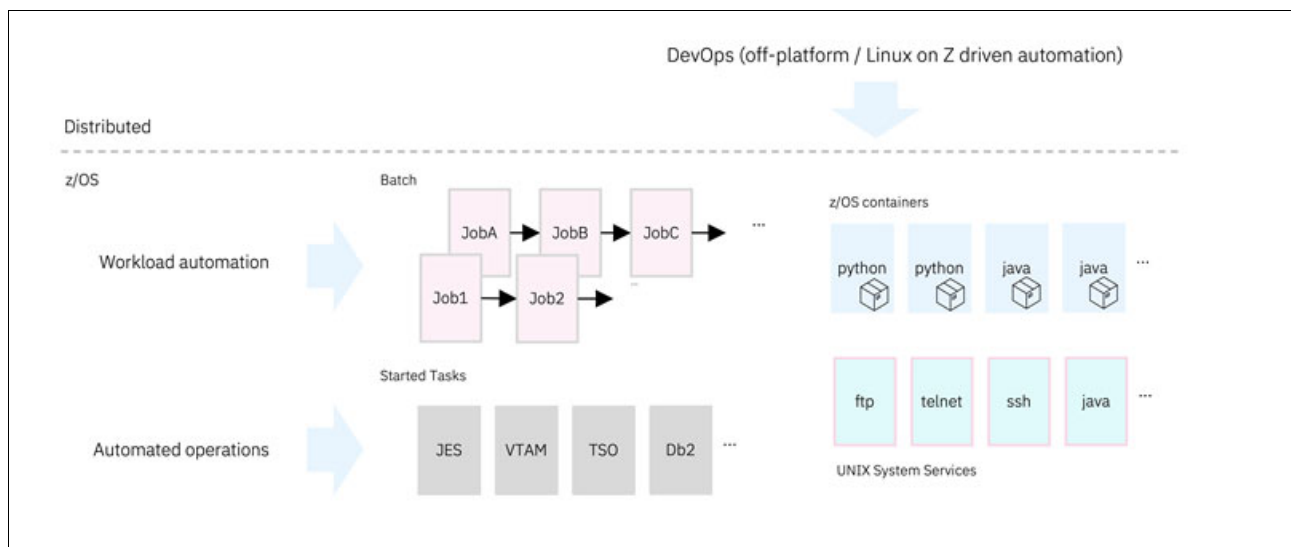


Figure 3-3 Automation landscape

Automation can be triggered in various ways. The following list highlights common triggers, presented in no particular order:

- ▶ Activity following incident remediation

Operators perform these steps to resolve an incident after identifying the root cause and when predefined automation is available. Examples include predefined commands, scripts, and automation requests that can be started from the z/OS console, a chat window, or submitted as an Ansible playbook. Automation can also be triggered by predefined or custom skills that are implemented in any form and offered by IBM watsonx Assistant® for IBM Z.

- ▶ Ad hoc requests for the operations team

Although incidents are the most common reason for ad hoc automation, requests might also occur during normal operations, for example, to start an application that is needed only under specific conditions.

- ▶ An event

This form of automation does not require human intervention. When an event has a clear symptom and a known, finite set of possible responses, it can be addressed directly through automation. An example is a Write-To-Operator-with-Response (WTOR) message that indicates the current SMF data set is full. AO responds to this event by switching to a new, empty SMF data set and unloading the previous SMF data set into a sequential data set, which can then be processed and archived by other tools.

- ▶ Planned outage

Operations sometimes stop to perform application, system, or data center maintenance. During these periods, business workloads cannot run at all or run only with restricted capabilities. There is strong pressure to minimize these times, which requires a high degree of automation to help ensure that applications and systems shut down in an orderly manner and restart quickly after maintenance is complete. Mainframe customers perform these activities during a maintenance window.

- ▶ Unplanned outage

This automation is similar to what is described under planned outage, except that an ordered shutdown is not possible and specific recovery steps must run before the application or system can restart. These recovery steps require more specific and elaborate automation. Rather than implementing and testing this automation yourself, you can rely on the vendor to provide a robust and highly automated solution. IBM offers Geographically Dispersed Parallel Sysplex (IBM GDPS®), which is a service that is built on IBM Z NetView and IBM Z System Automation that enables different solutions for availability and disaster recovery (DR).

- ▶ Schedules

Beyond maintenance windows, countless schedules determine when certain activities must occur and what conditions must be met before other activities can begin. The most common schedule is the daily plan in IBM Z Workload Scheduler, which describes all applications that are prepared by the job preparation team to run that day. These applications represent the business workloads that run throughout the day or during the batch window. The daily plan specifies the order in which these applications run, when and where they run, and which ones must finish by a certain time.

3.4.2 History

Two decades ago, tools such as Ansible and Kubernetes did not exist. However, AO and job scheduling were core topics in IT operations. Many concepts, such as using a declarative description of automated tasks, were implemented differently by using the capabilities that were available at the time. These concepts are effective and continue to influence modern automation tools.

AO began more than 40 years ago, around the time personal computers were first introduced. By the early to mid-1980s, it became clear that operators could not continuously monitor the system console to manually respond to events. During this time, AO emerged to filter messages for operators and introduce actions in response. Later, abstractions for starting and stopping workloads simplified how operators interacted with those workloads. Instead of remembering specific parameters to start or shut down a workload, operators could use a start or stop command that looked the same for any workload. The automation team prepared the specific parameters in advance as part of the automation configuration.

Similarly, job scheduling originated in the 1980s, when the rise of batch work required better organization of numerous batch jobs. The primary goal was to complete the daily batch workload within the limits of the available system capacity. From a business operations perspective, it became practical and necessary to plan batch job flows, understand dependencies, and optimize run time so that all workloads finished on time. The discipline of operations planning and control also influenced IBM's first product in this space.

Both automation disciplines evolved, modernized, and simplified over the years, but their core functions and capabilities remain mostly unchanged. Organizations and users build ecosystems around these products by using tools or custom interfaces to adapt them to specific operational environments. You, especially if you are part of an operations or automation team, should look for opportunities to replace manual processes with modern automation and integrate them with existing capabilities whenever possible.

3.4.3 Automated operations

As the name suggests, AO aim to create IT operations that are highly automated and largely free of human intervention, except when automation is unavailable or decisions require human judgment. Typical AO tasks include starting and stopping z/OS started tasks, performing an automated start (an initial program load (IPL)), shutting down the system, and recovering from failures. Other tasks include modifying the CPU and weight characteristics of IBM Z logical partitions (LPARs), activating temporary capacity, and activating or deactivating LPARs in response to known or unknown changes in workload demand or for DR purposes.

The most important capabilities of AO are described in the following subsections.

Message automation

AO evolved as an automation discipline from monitoring and automating the many messages that are issued by z/OS, middleware, and applications to the z/OS console. These WTO messages were historically the only mechanism for programs to communicate with operations and are still used today. The operating system defines how these messages are processed and how programs operating at the z/OS subsystem level can intercept them. Therefore, messages serve as an important event source on z/OS.

Messages should follow the z/OS message standards that are described in [z/OS MVS System Messages Vol 1](#) (compliance is not enforced). Messages begin with a unique message ID, which ends with a suffix indicating the message type. After the message ID, the program can include human-readable text that may span multiple lines. Each message also includes a descriptor code that has a standardized association with the suffix of the message ID.

If you follow these message format practices, it is relatively straightforward to filter and react to messages. However, the reality is different, which is why message automation became sophisticated. Because message IDs are not always unique (and sometimes cannot be), it is often necessary to extract content from the message text to decide what to do with a message. Some applications do not follow the practice of creating proper message IDs with the correct suffix or use inadequate descriptor codes, which prevents a generic reaction (for example, treating all eventual action messages ending with suffix E in the same way). Therefore, it is necessary to create message rules to trap and respond to individual messages.

As described in later subsections, messages also reflect application state. When two applications depend on each other, the second starts only after the first is running. AO uses status messages to determine application state and control which application to start or stop next.

Non-WTO messages

WTO messages are one of the primary ways that applications communicate with operators. Often, messages are written only to the application-specific job log in the JES SPOOL. Many applications run as UNIX System Services processes and write their messages to terminals or to files that are associated with the UNIX `stdout` file descriptor. These messages often also end up in the JES SPOOL.

The content and format of these messages are determined by the application. Often, applications that are originally developed for z/OS follow WTO message formatting practices for job log messages, but not always. A common log format, often used in modern distributed applications, is the exception rather than the rule.

AO must abstract the type and origin of messages to apply a consistent set of rules and actions across all message types.

Handling disruptions

Most messages are informational and end with the suffix “I”. However, informational does not mean ignorable because some messages can be interpreted as warnings or even errors. In other cases, messages include a descriptor code that indicates a critical or immediate action, requiring the operations team to respond.

Regardless of severity, AO is expected to respond based on the nature of the message. If an application fails, AO can attempt a restart. In more severe cases, the application might be restarted on another system (see “Starting and stopping”). If resource states change, for example, when an SMF data set becomes full or JES SPOOL usage is high, AO can automatically issue a corresponding reply or command. The response depends on how each installation chooses to handle it.

Starting and stopping

Many operations activities are either planned or ad hoc. They should be deployed through the existing AO. Goal-oriented AO is where the wanted status of a resource is set and the automation strives for goal attainment.

On request, AO must be able to start individual applications and the z/OS software stack during an IPL. Similarly, single applications must be stopped through a request that orderly shuts down the system. AO must understand the start and stop dependencies, and every application must run the steps in sequence. Status messages (WTOs) determine when one application reaches its wanted status and inform the AO when it is time to act on the next application in the dependency chain.

Policy-based automation is where resources and their dependencies are described in a policy rather than in scripted code. The AO must also be neutral about what type of work it must handle. Operators can use a generic interface to start applications, such as started tasks or UNIX System Services processes. Although it is possible to submit jobs through AO, you should centralize all work that is related to a batch job within the job scheduling domain (see 3.4.4, “Batch processing or job scheduling” on page 28).

Ad hoc requests can be handled directly by operators from the AO user interfaces or programmatically through external requests, for example, by using OpenAPI REST APIs. Planned activities are often handled through schedules where the operations team can inform the AO in advance when and for how long a certain application should be available or not.

Often, it is not a single application that must be started, but a whole set of applications, for example, a Db2 subsystem with master, database services, distributed data server, and stored procedure address spaces. Especially when dealing with such situations manually, groups are an effective means to simplify the operations tasks. The ability to collect applications within a group then enables an operator to issue a single request instead of many individual requests.

Operator assistance

In addition to groups, another assistance function for operators is performing activities with a single command. Automation administrators prepare these commands in the form of scripts and document them. This approach reduces complexity when human intervention is required and makes interaction with the system more intuitive.

Another form of assistance is narrowing the scope of what must be monitored. Groups can aggregate the status of their applications and report that aggregated status as their own. The same concept can also be applied to other resources of interest to an operations team, such as critical messages, health checks, job scheduling resources, and tapes. If the individual status can be aggregated at the system level, operators can assess the overall status by examining the system and delve into details only when an issue exists.

Although not common, there are situations when SMEs must perform activities without the involvement of AO. In this case, it is convenient when the operations team can suspend automation for individual applications or groups. Any interaction with these suspended applications that normally lead to immediate reactions is then possible without risking unwanted side effects and without impacting the automation of the remaining system.

Other assistance functions are provided when operations teams must handle more complex commands, which can sometimes be a challenge even for system programmers. Through its user interfaces, AO can help operators understand coupling facilities and their structures and manage duplexing status. It can also help review and modify certain datasets. AO can gather IPL information and display a history of important members to spot changes. Also, it can help manage options for system dumps and SAN Volume Controller dumps and manage serviceability level indication processing (SLIP) traps.

High availability and disaster recovery

Advanced AO capabilities relate to high availability (HA) and hardware operations to help installations manage LPAR capacity and perform IPLs. Hardware operations are the most important for disaster recovery (DR). However, operating in a disaster scenario is the responsibility of IBM Geographically Dispersed Parallel Sysplex (GDPS) because of the specific orchestration that is required to help ensure proper failover and maintain data consistency.

Groups can be containers of resources that can be managed collectively through a single request. For HA, more advanced concepts are required. An HA group can be used proactively (planned) or reactively (unplanned) to move its members from one system to another. A server group can dynamically adjust the number of active members and perform member recycling in chunks, always striving for HA and avoiding service disruptions. For readers familiar with Kubernetes, a server group is similar to a deployment with a variable number of pods.

3.4.4 Batch processing or job scheduling

The term *batch* refers to all types of workloads that run in the background for some time without human intervention. In contrast to started tasks, these workloads are scheduled either according to a specific plan or as ad hoc jobs and stop at a defined point. When referring to batch processing or job scheduling, the term primarily means the planning and running of batch jobs that are specified by the MVS Job Control Language (JCL) and submitted to JES. However, the terms *batch workloads* and *workload automation* are sometimes used interchangeably, and these workloads often cross platform boundaries and include jobs running on third-party platforms.

Overview

This IBM Redbooks publication cannot replace a deeper study of job scheduling concepts. However, it highlights the most important capabilities in a concise form so that you can understand what matters in the context of ITOps. These capabilities align with the characteristics that analyst firms describe as Service Orchestration and Automation Platforms (SOAP).

Figure 3-4 on page 29 illustrates an enterprise workload automation solution that includes both distributed and mainframe systems.

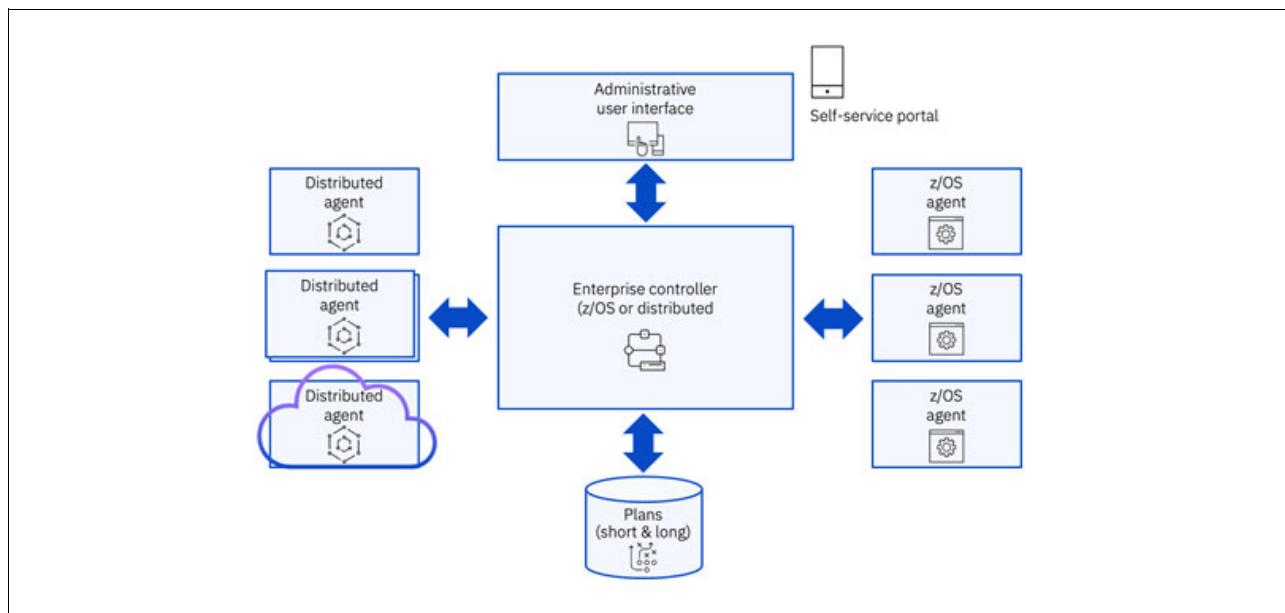


Figure 3-4 Typical workload automation solution

Workflow orchestration

One key prerequisite of any job scheduler is the existence of a plan. The scheduler must know what to schedule and when to schedule it. The plan usually considers the calendar, including weekdays, holidays, and special periods such as month-end processing. It also specifies the dependencies for each job. In addition to the specific date or time, other prerequisites can include jobs that must run successfully; specific hardware that is required (such as tape drives); and the availability of z/OS resources such as JES initiators that are assigned to a particular job class.

Despite planning, jobs can fail and must be recovered. With most jobs, manually monitoring the runtime status is unfeasible. The job scheduler, tightly integrated with JES and SMF, can monitor when jobs finish and how they finish. The work planner can instrument the JCL with specific statements that define how the job scheduler reacts if there is a failure. Typical actions include rerunning the job, submitting another recovery job, or issuing a WTO to alert the operations team about the incident.

For rerunning a job, the job scheduler must help ensure that the datasets and catalog entries that are created up to the point of failure are cleaned up so that the resource state is the same as though the job ran for the first time.

In enterprise environments, jobs are often grouped to form larger applications, and jobs and applications might have dependencies on each other. A job scheduler helps define these relationships so that a successor job or application runs only when its predecessor finishes successfully. Sometimes, this dependency can be expressed conditionally, depending on the predecessor's completion status and return code.

Jobs can run in parallel but might need to update the same data set. To help ensure correct serialization, a job scheduler must provide a mechanism for serialization, managing that data set as a logical resource. Other uses of logical resources include signaling between the job scheduler and other management software running on z/OS, such as CICS or AO.

Advanced workload orchestration helps work planners predict run time under certain conditions based on the plan. For example, if a dataset is delivered later than usual, the work planner can simulate how that delay impacts the running of downstream jobs or applications and adjust the plan to avoid missing a deadline.

Event-driven workload automation

There are use cases when workloads are automated outside of what is linearly specified in the plan. This approach is called *event-driven workload automation*. Do not confuse it with AO, where messages are monitored and responded to.

Events involve activities with datasets or file systems, such as when a data set is closed or a file system is changed. Another event can be the status change of a logical resource. In these cases, the job scheduler can respond by modifying a resource in the plan or by releasing a dependency on an operation that monitored a data set.

In other cases, jobs must run repeatedly, for example, every first day of the week or every third Friday of the month. Some of these periodic definitions might also include conditions, such as skipping the running of the job when the selected day falls at the end of a month. These use cases require the job scheduler to maintain a calendar as part of the long-term plan.

Finally, there are cases where an event is generated internally when the job scheduler detects a job that is too close to its expected deadline. In such cases, the job is considered to be on the critical path. Here, the job scheduler can respond by alerting the operations team or by interacting with z/OS to reset the job to a preferred service class.

Job scheduling and observability

Batch workloads typically consist of many steps and are often part of a hybrid application running on different IT platforms. Therefore, monitoring and observability principles apply to all workloads, including batch.

Modern programming languages and container orchestrators, such as Kubernetes, provide built-in support for integration with observability frameworks like OpenTelemetry. This capability minimizes and sometimes eliminates the effort that is required to instrument applications for observability.

For batch workloads, the job scheduler is responsible for providing this instrumentation. Transparent to the applications, the job scheduler shows metrics, logs, and traces about the application and its constituent jobs and job steps in an open format to the observability platform of choice. There, the end-to-end flow can be visualized, and progress can be monitored alongside other monitoring data. This observability solution complements the job scheduler's own user interface, which is required for administering the plan and running the batch workloads.

Analytics and applied AI

The growing volume of batch workloads increases the complexity of managing resources and dependencies among numerous jobs and applications. Concurrently, the operations team must handle this workload with limited IT resources and help ensure that batch results meet business deadlines. In this context, applying methods from the field of AI is a way to identify optimization opportunities or detect anomalies during run time for early warning.

In 3.5, “The role of artificial intelligence in ITOps” on page 31, you learn about these capabilities in more detail, with a dedicated section on AI and how it can help simplify operations, including batch processing.

Data pipeline management

With batch workloads increasingly distributed, how do jobs send or receive data between agents? A modern workload automation solution must provide a secure, fast, and reliable way to transfer data between its agents and across platforms without requiring an extra third-party component.

Batch workloads often require integration with data lakes, so data pipeline management must also include the ability to integrate with big data through extract, transfer, and load (ETL) processing. In an ever-growing and evolving ecosystem, a flexible workload automation solution offers plug-ins that customers may use to integrate their big data tools into enterprise workflows.

Self-service automation

To provide the most value for the lines of business, a workload automation solution must align with the needs and experience of its users. Although the solution involves technical complexity, you do not need to be exposed to it.

A modern workload automation solution offers a catalog of prepared business workflows through a self-service portal, where you can select services as needed. The solution adds the selected service to the plan, orchestrates the necessary steps, monitors the process, and fulfills the service.

For more information, see Chapter 5, “Where to start” on page 63.

3.5 The role of artificial intelligence in ITOps

This section focuses on how AI can augment existing capabilities to better manage an ever-growing volume of data and identify the critical data points that require the attention of operators and SMEs. It also explains how large language models (LLMs) in the context of operations can help guide users in solving day-to-day tasks more productively.

AI can be categorized into different forms:

- ▶ **Assistive AI:** Supports users by surfacing insights, simplifying tasks, and enhancing understanding. This form of AI is found in various assistants that are built into or integrated with day-to-day tools.
- ▶ **Predictive AI:** Anticipates issues or needs to guide planning and prioritization. This form of AI is not limited to assistants and is present in many tools.
- ▶ **Agentic AI:** Takes the initiative and acts autonomously to drive outcomes. It is the most advanced form of AI, using the power of LLMs to perform reasoning and acts based on the user's intent.

3.5.1 Brief history and evolution

AI is the ability of a computer to learn, reason, perceive, and decide similar to a human. In 1959, IBM researcher Arthur Samuel coined the term *machine learning* (ML) when working on the problem of how computers can learn to solve tasks without being explicitly programmed.

ML is a subset of AI that is focused on creating generalizable predictive patterns by using supervised or unsupervised learning. *Supervised learning* trains a model to compute outcomes based on labeled data, where each input has a correct answer (*label*). The model learns from these examples to make accurate predictions on unseen data. *Unsupervised learning* trains a model to create clusters based on unlabeled data. During inference, such a model can predict which cluster that the new data belongs to, even if it has not seen that data before.

Deep learning is a subset of ML that uses various types of neural networks and is trained with massive amounts of data, enabling advanced use cases such as computer vision. *Neural networks* are inspired by information processing and distributed communication nodes in biological systems, particularly the human brain.

From deep learning, the evolution continued to foundation models. *Foundation models* are large deep learning models that are trained by using self-supervised learning on vast amounts of data, such as billions of pages of text from the internet. Equipped with this extensive knowledge, a foundation model can be applied to a wide range of use cases, such as generating or summarizing text. An LLM is a foundation model that specializes in generative AI that is applied to text.

The following sections describe where AI can be used in ITOps.

3.5.2 Anomaly detection

Detecting anomalous behavior in the IT environment is a common method to prevent unforeseen disruptions to business workloads. While anomaly detection often implies ML, in this context it is broader and not limited to ML applications:

- ▶ **Manually defined static criteria or thresholds:** This approach involves manually set boundaries, such as those found in OMEGAMON situation events or monitoring for specific message identifiers. The boundaries define what constitutes alertable behavior.
- ▶ **ML approaches:** These approaches involve training models on historical data to recognize typical patterns. The model evaluates new data against these learned patterns to determine the degree of match. Examples of ML algorithms include clustering, classification trees, random forest, and deep learning methods. To increase prediction accuracy, ML models can be combined to achieve higher overall confidence. Such a combination is referred to as an *ensemble*.
- ▶ **Statistical techniques:** Like ML, these techniques learn from historical data to identify patterns, but they use statistical methods rather than ML algorithms. These techniques assess incoming data for pattern matching. Examples include standard deviation, interquartile range, and Autoregressive Integrated Moving Average (ARIMA).
- ▶ **Custom algorithms:** These algorithms are tailored to the specific domain or based on historical experience with the data. They use expert knowledge to determine what qualifies as an anomaly.

While ML and statistical techniques might seem interchangeable, debates exist regarding their distinction, particularly in the context of anomaly detection. Despite these differences, both rely on mathematical or statistical principles and often yield more robust results when used together. In practice, anomaly detection frequently involves a multi-pronged approach that integrates various techniques to effectively address real-world problems.

Learning-based anomaly detection versus static thresholds

Static thresholds require users to define what constitutes abnormal behavior and create corresponding rules. When the threshold is known, it serves as an effective monitoring mechanism, often outperforming data-driven learning methods because of their inherent uncertainty. However, static thresholds have limitations:

- ▶ Seasonal variations (for example, time of day, day of week, or end of quarter) might require fine-grained, time-consuming, and error-prone threshold definitions and adjustments.
- ▶ Evolving thresholds as business workloads change, software is updated, and infrastructure is modified.
- ▶ Single-metric limitations where thresholds based on one metric or message identifier can be restrictive, and rules involving multiple criteria or metrics are more complex to manage and implement accurately.
- ▶ Limited coverage because enterprises often maintain only a small set of static, simple thresholds, potentially overlooking unforeseen issues (“we don’t know what we don’t know”).

Static threshold-based methods, although straightforward to implement, often lack the flexibility that is needed to adapt to dynamic conditions and might miss unknown situations. *Dynamic threshold methods* learn from data and address many limitations of static thresholds but are subject to uncertainty stemming from the algorithm and imperfect data. Ideally, the most effective strategy combines both approaches, and is tailored to the specific context and requirements.

Predictive analysis

How predictive is predictive analysis? Predictive analytics in ML often rely on trends or patterns to forecast future values. Here are two approaches for predicting situations in ITOps and what needs to be considered:

- ▶ Using historical data: Mathematical formulas can estimate future metric values based on trends. These forecasts can range from a few minutes to several months, with accuracy decreasing as the prediction horizon extends. The optimal prediction window should balance minimizing false positives with the use case and required reaction time.
- ▶ Monitoring subtle anomalies: Computers can detect anomalies that might go unnoticed by humans. If the situation continues to develop, more components or transactions might be flagged as abnormal (increased quantity), and anomalies might occur more frequently. Triggering alerts on the first anomaly can lead to numerous false positives, while delaying alerts too long might prevent timely intervention.

The predictiveness of analytical methods can be measured by comparing the alert time with the time of impact that is noticed by the user or detected by traditional monitoring tools. Predictiveness depends on the specific problem scenario and available data. Certain issues, such as ones that are caused by a rogue employee unplugging a power supply, cannot be predicted by using ITOps data. Also, some symptoms escalate so quickly that there is insufficient time for proactive prediction.

Understanding false positives

One common expectation for anomaly detection is to identify potential IT system issues. However, drawing a direct causal link from detected anomalies to application or system failures involves numerous assumptions and a complex chain of reasoning. Users often exercise subjective judgment to determine whether the perceived causal relationship constitutes a false positive.

When an anomaly surfaces in the mainframe environment, it might resolve itself through existing automation or built-in resilience in the operating system, subsystems, or sysplex, minimizing impacts on workloads without human intervention. Users who are unaware of these automated responses might view such notifications as false positives. Conversely, informed users might dismiss these alerts as unnecessary noise. If an alert is omitted because of an automated response, there is always a risk that the automated response might fail to correct the issue.

How to get a good return of investment from anomaly detection

Learning-based anomaly detection and its predictiveness inherently have limitations, often depending on specific problem scenarios and user expectations. False positives might be confirmed retrospectively, requiring more work and time to investigate abnormal symptoms. However, using learning-based anomaly detection to establish dynamic criteria is essential because of the complexity of workloads and the dynamic nature of business environments.

Enterprises can optimize the usage of anomaly detection technology and maximize return on investment by using the following approaches:

- ▶ Proactive problem identification: Using anomaly detection as a predictive tool to anticipate and address potential issues before they escalate.
- ▶ Post-incident analysis: Using anomaly detection to pinpoint probable problem areas or uncover root causes after an impactful event.
- ▶ Resilience improvement: If anomalies do not lead to impactful issues, they can serve as signals to refine and enhance IT environment resilience. This method might involve refining system configurations and creating more automation for the symptom. If these minor or moderate anomalies *are* false positives, they can be explicitly disregarded through a user feedback loop.

3.5.3 Generative AI

When you refer to generative AI, the term *generative* means generating text by using an LLM. An LLM is trained on a large corpus of text so that it can understand sentences and generate new sentences with correct grammar and spelling. An example of an LLM is IBM Granite® 3.3 8b-Instruct, which is designed for text generation.

To interact with an LLM, you must provide a prompt. A *prompt* typically consists of instructions that are followed by the user query. This combined text forms the initial context that is passed to the LLM. Before processing, the words in the input text are tokenized. Each word is represented by one or more tokens. The LLM uses this context to generate tokens based on the instructions. Each generated token is added to the context to predict the next token. After the sentence is complete, or the requested number of output tokens is reached, the LLM returns the response.

Usually, an AI application encapsulates this process. You interact with the application through a chat interface or a field where you enter your prompt and submit it. The AI application performs the steps and returns the response that is generated by the LLM.

Retrieval augmented generation

Although LLMs are general and have broad knowledge, they are not all-encompassing. Because a company's operations handbook is unlikely to be part of an LLM's training data (unless the company trains its own LLM), the model might not know how to respond to a user prompt. It responds, but the answer is likely incorrect. Another issue is that information becomes outdated. For example, if the LLM was trained with knowledge from z/OS 3.1, it might not answer questions about z/OS 3.2 because the information is missing or obsolete.

Retrieval augmented generation (RAG) is a technique that addresses both issues. Instead of training an LLM with these documents, the documents are divided into smaller chunks and stored in a vector database. For each chunk, an embedding vector is calculated to represent the content in an N -dimensional space, where N is typically large. The IBM Granite embedding model, `granite-embedding-30m-English`, for example, uses a dimension of 384.

When you ask a question, RAG computes an embedding vector for your prompt. It then searches the vector database for the top N chunks whose vectors are most similar to the prompt vector. Vector databases use algorithms from the class of approximate nearest neighbor algorithms to perform this task. The matching chunks are returned and added to the original prompt. Both the chunks and the prompt are then passed to the LLM to generate the response.

Applying RAG improves the accuracy of LLM responses. By instructing the LLM to use only the documents that are returned by RAG, you can reduce hallucinations. If the documentation becomes invalid, you can update the vector database without retraining the LLM.

RAG is often used when users interact with an assistant to ask general or company-specific questions based on information that is stored in documents in a vector database.

Assistants and agents

An *assistant* is a type of AI application that provides a way for users to chat. An assistant can include functions or capabilities that go beyond interacting with the LLM. To determine which capability to use, the assistant performs *intent detection*, where an LLM infers the most likely intent based on the user's prompt.

One capability might be the ability to perform RAG queries. In this case, the assistant uses RAG to find documents that match the user's prompt. Then, it interacts with the LLM, passing the retrieved document chunks along with the prompt to generate a response.

Another capability might be running a tool. For example, a tool might be a command that displays the current z/OS LINKLIST. In this case, the assistant attempts to find the tool that best matches the user's prompt. When it finds a matching tool, the assistant might engage the user to provide more parameters, such as the name of the system where the command should run. Then, the assistant runs the tool and returns the result, such as a list of datasets in the LINKLIST.

Another capability is running an AI agent. Unlike a tool, which typically follows a single-step process (prompt and then response), AI agents use multi-step processes. They interact with the LLM multiple times to determine the next action, perform that action, and evaluate the result. The agent decides whether the result satisfies the user's initial prompt. If not, it repeats the process. These agents are called *ReAct agents*, short for reason and act.

Multi-step processing raises questions about how an AI agent reasons, selects a tool, and interprets results. AI agents address these questions by providing transparency. They inform the user about the steps that are taken and the reasoning behind each decision. This transparency builds trust.

More applications are likely to adopt assistants because they help automate tasks that users would otherwise perform manually. For example, you can submit a batch job to reallocate a partitioned data set in response to a B37 abend by using only natural language.

Simplifying mainframe operations

Chapter 3, “Artificial Intelligence for IT Operations capabilities” on page 9 described general capabilities that enable modern mainframe operations without detailing specific products. It explained the characteristics of these capabilities and why they are important.

This chapter focuses on solutions that simplify operations, as shown in Figure 4-1.

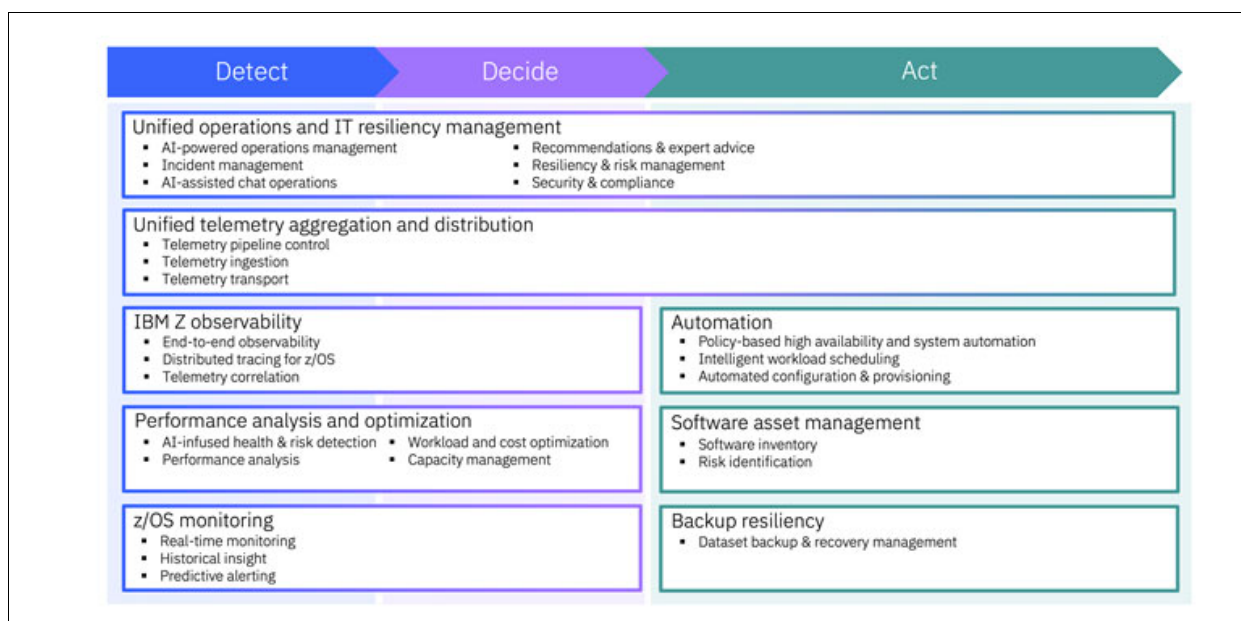


Figure 4-1 IT operations capability map

4.1 Single point of control for IT operations

This section focuses on IBM Concert® for IBM Z, which addresses business resiliency from a single interface.

Chapter 2, “Challenges” on page 3 explained that mainframe operations teams face challenges due to the large volume of data that is generated and presented to operators and subject matter experts (SMEs). This volume often creates more noise than insight.

To improve productivity, these roles need a holistic overview of workloads running on the mainframe and actionable insights that are derived from the data. When incidents occur, visibility and insights help analysts quickly focus on what matters. Analysts also need support in the form of guidance or automation that helps them remediate the incident confidently and restore service as quickly as possible.

Incidents cannot be avoided. However, when incidents occur, focus and speed are critical. Mainframe teams also value early detection of potential issues so that they understand the possible risks that these issues present. This early visibility enables them to stay ahead of the curve and remediate issues proactively before they become actual incidents.

Finally, because the infrastructure and applications running on it are constantly changing, mainframe teams must help ensure that all guidelines regarding compliance and security are met. If they are not met, the highest priority is to mitigate those risks proactively to protect data and avoid service disruptions that are caused by failures or physical attacks.

These challenges can be overcome by a set of capabilities in three dimensions: operations management, resiliency and risk management, and security and compliance management. These dimensions are discussed in the following sections.

4.1.1 Operations management

The operations management dimension primarily addresses the lifecycle of an incident from inception during the Detect phase; through isolating the issue and performing root cause analysis during the Decide phase; to remediating the issue and restoring service during the Act phase.

IBM Concert for IBM Z is the hub of IT operations (ITOps) on IBM Z. It uses artificial intelligence (AI) to detect and prioritize anomalies based on an understanding of normal behavior and raises events to inform operations before anomalies can impact service levels. Furthermore, events are correlated and grouped by temporal and topological relationships to reduce noise and draw the user's attention to what matters most. Topology visualization from hardware to system to middleware, including transactions, helps during analysis and enables operators to understand the impact of event groups. Finally, IBM Concert for IBM Z provides AI-generated recommendations and enables operators to interact directly with z/OS to resolve incidents.

With these capabilities, IBM Concert for IBM Z simplifies how support staff operates the mainframe. The entire user experience is designed to provide all the necessary functions through a unified, intuitive dashboard interface, eliminating the need to switch to another tool. IBM Concert for IBM Z also helps reduce mean time to resolve and enables operators to act with confidence to resolve issues quickly and proactively.

The following sections illustrate these capabilities in more detail.

Data sources

IBM Concert for IBM Z supports data ingestion through the Common Data Provider (CDP) and optionally through the IBM OMEGAMON Data Provider (ODP) (for more information about both tools, see 4.2, “Observability: Mainframe scoped and across the enterprise” on page 45).

In addition, IBM Concert for IBM Z adopts an open standards approach by facilitating integration with third-party solutions through an OpenTelemetry receiver to ingest any data that conforms to the OpenTelemetry specifications for metrics and logs.

Intelligent anomaly detection

Rather than relying on manually tuned thresholds, intelligent anomaly detection identifies anomalies by comparing current key performance indicators (KPIs) with historical values. Reducing false positives is a constant challenge, which is why different algorithms from statistical methods and machine learning (ML) are used together to maximize prediction performance.

Training and inference occur continuously in the background, transparent to users. The KPIs cover a broad range of z/OS and its middleware subsystems, such as Db2, IBM Customer Information Control System (CICS), IMS, and IBM MQ. Also, observations from the z/OS SYSLOG, such as messages that require operator attention or appear more frequently than usual, are also analyzed.

Figure 4-2 shows a KPI (the line) and its typical value range (the shaded area). You can identify the outlier, which is automatically graded to determine the severity of the anomaly.

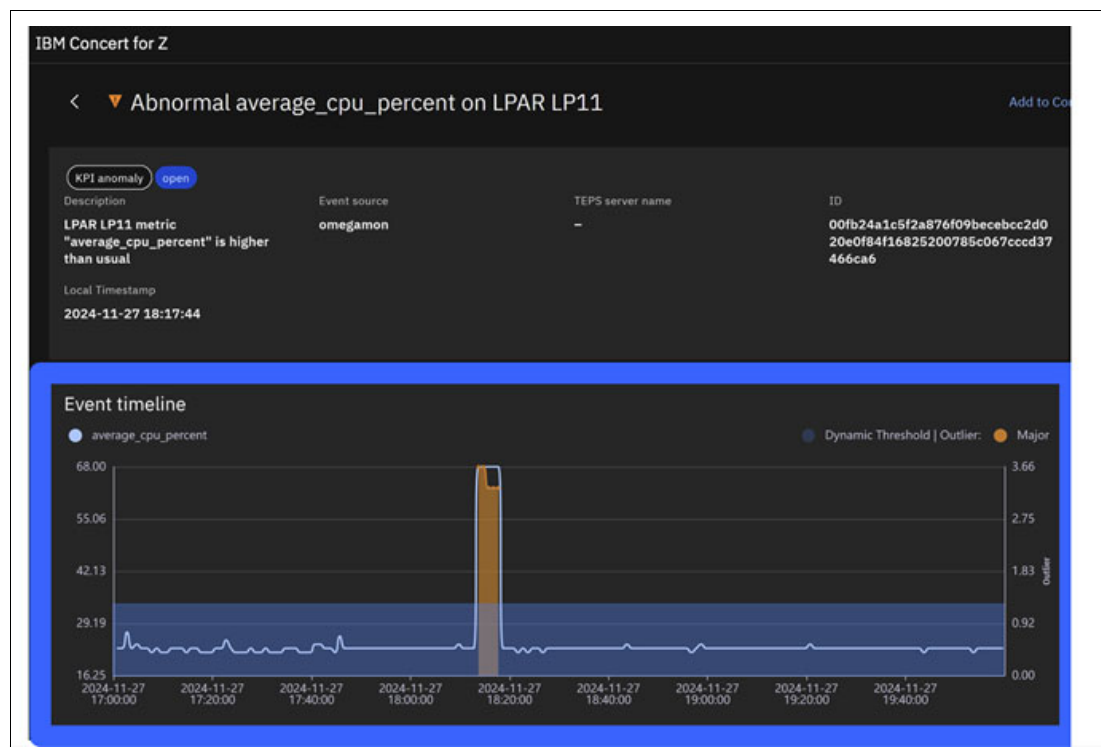


Figure 4-2 Anomaly detection

Automated event correlation

Many events can occur and create substantial noise unless they are examined and grouped in a way that makes sense to the operator. The system groups events based on two assumptions:

- ▶ Events are likely correlated when they occur at approximately the same time.
- ▶ Events are likely correlated when they are generated on behalf of related configuration items, for example, a CICS region running transactions that perform Db2 queries.

Although the system aggregates individual events, it also adjusts the severity of the resulting group based on the severity and frequency of those events. You can easily identify the most critical event groups and focus on resolving the issue (see Figure 4-3).

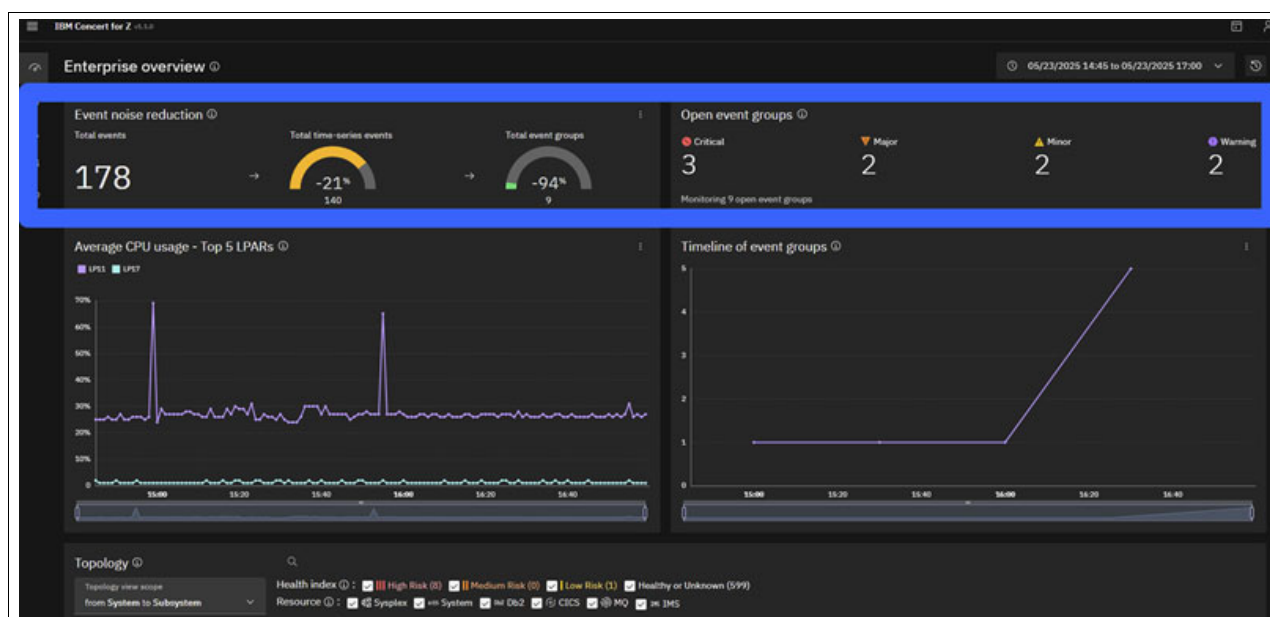


Figure 4-3 Event correlation and grouping

Centralized and context-sensitive dashboards

Operators and SMEs work most productively when all information is immediately accessible. They can access this information through a dashboard experience that provides both summary and detailed views within a single interface.

Users can drill down, move sideways, and move upward as needed, based on their current position in the analysis process. To guide navigation, users can reference the built-in topology (see “Transaction-level topology mapping” on page 41).

Metrics from System Management Facility (SMF) records or OMEGAMON monitors cover z/OS and its subsystems (Db2, CICS, IMS, and IBM MQ) and are presented in context-sensitive dashboards. Examples are shown in Figure 4-4 on page 41.

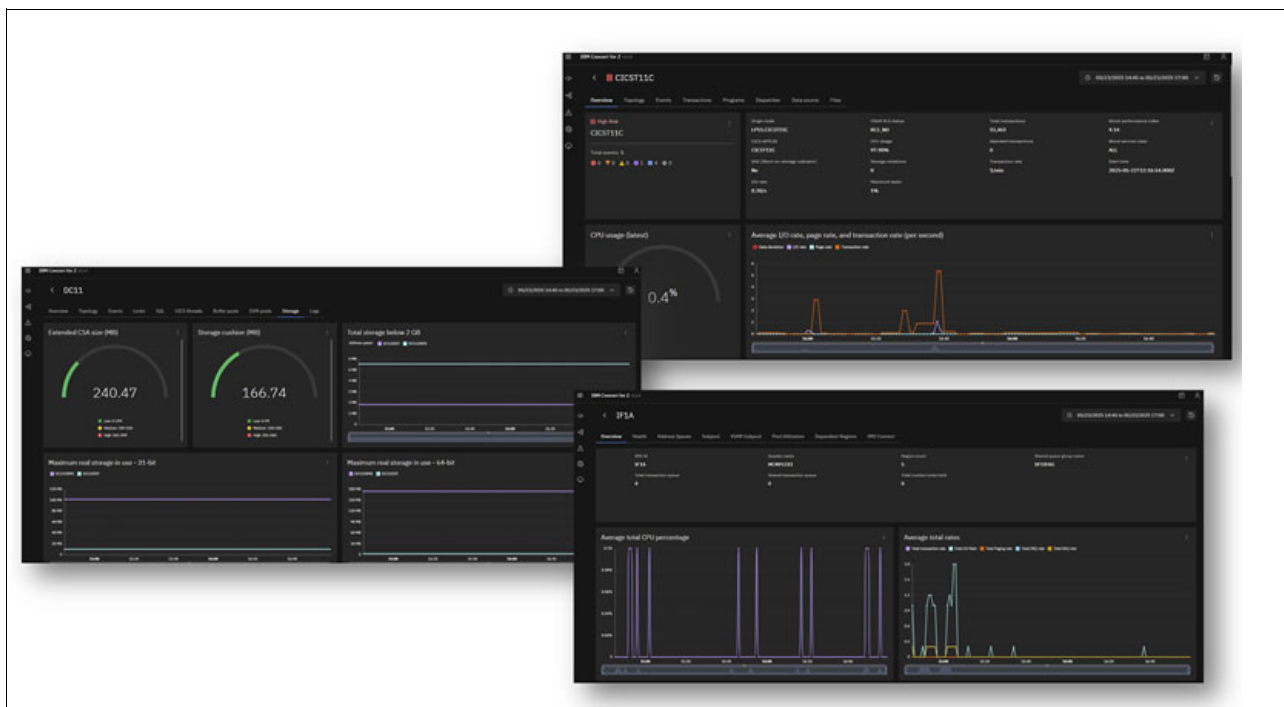


Figure 4-4 Context-sensitive dashboards

Each dashboard consists of a set of sub-dashboards where related metrics are grouped to organize content and help users digest the information more effectively.

Transaction-level topology mapping

One use of topology is to understand the context of a configuration item in focus. For example, users are not viewing a CICS region in isolation. Concurrently, they can see whether events are related to it and what their severity is. They can also see which system it is running on and which other subsystems are referenced by its transactions. This approach provides operators and SMEs with a holistic understanding of an issue, clearly showing how mainframe resources relate to each other.

Another use of topology is for navigation, as shown in Figure 4-5. When users start at a high level, viewing only critical systems, they can click specific nodes in the topology to focus on configuration items of interest. They can expand the detail level of the topology to see programs, transactions, queues, and other elements, depending on the type of the configuration item.

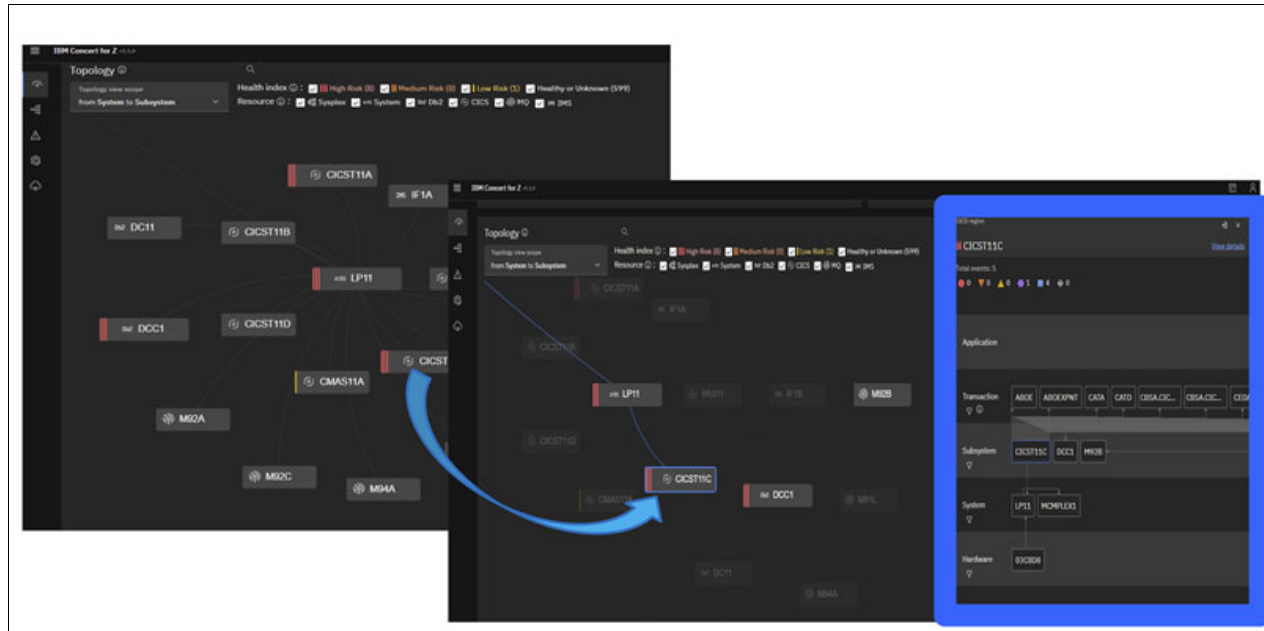


Figure 4-5 Topology for context and navigation

Getting expert advice

Regardless of how intuitive the dashboards are, subject matter details cannot always be hidden. Sometimes, experts are still required to understand and interpret the data.

One method to provide advice is to evaluate the anomaly and the associated KPI. For most events, expert advice exists and is matched against the observation. This advice is based on knowledge from IBM technical consultants and developers and is automatically presented while you view event details.

With AI, and its specialization that is called generative AI, expertise can also be provided through an AI chat to anyone seeking advice (see Figure 4-6 on page 43). With integration with watsonx Assistant for IBM Z, you can retrieve guidance in the form of questions and answers. For example, an early tenure operator or SME can ask domain-specific questions such as “What is the difference between MVS Busy and LPAR Busy?” when confronted with both metrics and wondering how to interpret them.

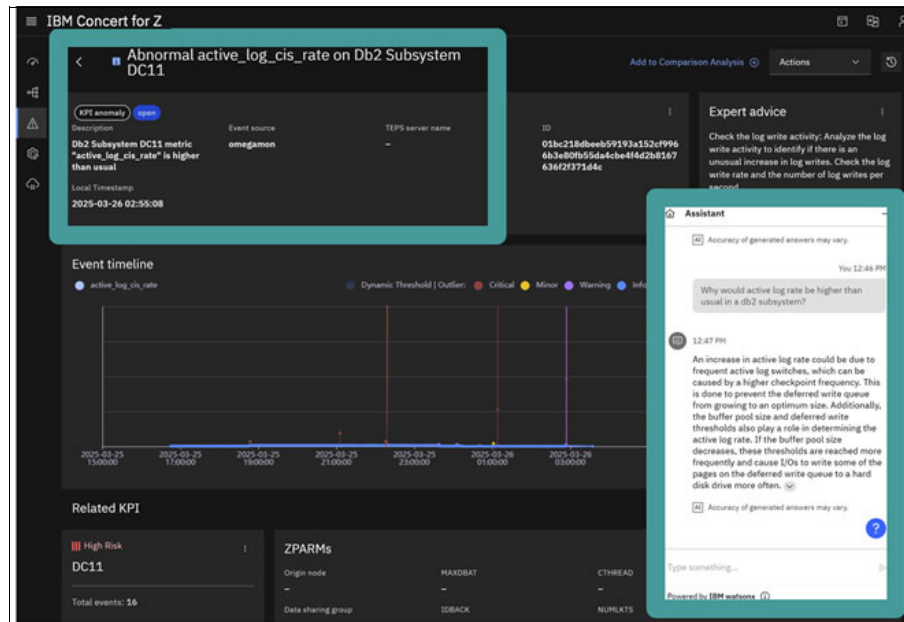


Figure 4-6 Getting recommendations from AI chat

Beyond technical questions in the IBM Z domain, you might want to ask for company-specific guidance about how the issue should be handled. Such proprietary operation manuals are tailored to the needs and procedures of each enterprise.

Finally, you can use the AI chat to request runtime information from the systems of interest or to run actions for automation. Examples include starting a command, running a script, or running any other form of runbook. IBM watsonx Assistant for IBM Z calls these automations *tools*, which can be selected based on user needs.

At the time of writing, IBM laboratories worldwide are working on the next step in the generative AI movement, called *agentic AI*. Unlike tools, agents evaluate your input, plan actions by using available tools or expert agents, and formulate a response for you. You can expect more capabilities to emerge in the coming months.

IT service management integration

Throughout the entire process (from detecting an issue, through root cause analysis, to remediation), you must create an incident ticket to use the incident management process. Use this ticket to record symptoms, track analysis steps and their results, and log the actions that are taken to resolve the issue.

Many mainframe clients use ServiceNow as an IT service management (ITSM) solution for managing the lifecycle of tickets that are created as part of incident, change, and problem management. Because this bookkeeping is necessary and serves an important purpose, it makes sense to integrate the ticketing system directly into your workflow.

For event groups, you can use the ServiceNow integration to create a new incident ticket. The ID that is created for the ticket is stored with the event group so that other users know that a ticket exists and can refer to it by ID to update the ticket status as necessary.

Integrated command console

During analysis, for example, when you must issue a z/OS DISPLAY command to check the current LINKLIST, or when a remediation plan exists, operators and SMEs must be able to access the system to perform these actions. Rather than using another tool, you can run actions directly from the built-in command console that is based on IBM Z NetView, from which you can run any command (z/OS or NetView).

This command facility enforces user authentication and acknowledges authorization through the System Authorization Facility (SAF). When you use TSO user ID credentials, you must authenticate before issuing commands. Every command is checked for the necessary authorization by using the existing SAF permissions that are assigned to the user.

4.1.2 Resiliency and risk management

Resiliency and risk management focus on assessing health insights for the IBM Z infrastructure and identifying risks that are associated with maintenance gaps. These insights help enterprises remain resilient in the presence of risks that are otherwise difficult to detect.

Visualizing and managing IBM Z infrastructure health

IBM Z IntelliMagic Vision for z/OS is used by performance analysts and capacity planners to gain a holistic understanding of the performance and resource consumption of the IBM Z infrastructure, and the z/OS and middleware running on it.

To support resiliency and reduce risks, the product provides various health insight reports about usage and possible exceptions across multiple categories, including IBM Z hardware, the I/O fabric, storage control units, devices, sysplex, z/OS Workload Manager (WLM), and middleware. The exception ratings are derived from best practices that are documented in IBM Redbooks publications and are largely based on the experience of IBM consultants.

Analysts can use these reports daily to assess the situation and identify possible areas for improvement. Although flagged exceptions do not require immediate action, they serve as early indicators of potential resource bottlenecks that can lead to service degradation or even outages if ignored.

Presenting this information at regular intervals (by default, every 24 hours) within IBM Concert for IBM Z, and providing automated risk scoring according to customer-specific measures, provides operators and SMEs with a holistic understanding of resource usage across the entire IBM Z infrastructure. This approach helps them better understand the overall risk. The related views are presented next to the dashboards for operations management.

Intelligent APAR management

Software maintenance is essential. However, changing software always introduces new risks and usually results in temporary service disruption during planned maintenance intervals. Therefore, mainframe clients should carefully balance the need for software maintenance with the potential impact of rolling out software into an existing environment. Clients need a more intelligent approach to managing software upgrades that enables system administrators to understand which software components are affected by missing maintenance and which z/OS environments are impacted.

With IBM Z Software Asset Management (IZSAM), the z/OS Management Facility (z/OSMF), and the Ansible Automation Platform, this solution helps you determine whether the software level across all z/OS environments is up to date or whether maintenance drift requires updates. The solution computes risk due to missing critical APARs and their prerequisites and relates that risk to the software that is installed and running on the various z/OS systems in the enterprise. The dashboards visualize the risk and graphically display the blast radius, showing the potentially affected environments.

With the help of watsonx, the solution can recommend a strategy for risk mitigation. It can automatically create change tickets based on risks and assign them to the appropriate maintenance focal point or even to an AI agent for the necessary software update steps.

4.1.3 Security and compliance management

For security and compliance management, IBM Concert for IBM Z focuses on managing certificates that are used across z/OS. The solution evaluates the health of installed certificates based on their age and expiration dates and provides dashboards to visualize this information. It includes automation to fetch current data from the security database and to initiate a change management workflow for creating a certificate, if needed. The automation is delivered as playbooks on the Ansible Automation Platform.

The solution integrates with various tools for handling updates. In addition to ServiceNow, it offers integration with GitHub, Jira, and others. When a ticket is approved, playbooks create a certificate and refresh it in the locations where it was previously installed. These capabilities streamline the certificate management process and help minimize outages and security risks.

4.2 Observability: Mainframe scoped and across the enterprise

This subsection explains how the IBM Z Artificial Intelligence for IT Operations (AIOps) portfolio supports IBM's observability strategy within z/OS, across Linux on IBM Z, and throughout the enterprise to address the growing complexity and volume of heterogeneous workloads running on different infrastructure tiers.

4.2.1 Telemetry aggregation and distribution

Near real-time distribution of monitoring data and logs from z/OS to multiple targets is supported by IBM Z CDP. CDP is part of IBM Z Operational Log and Data Analytics and the IBM Z Integration for Observability suite.

CDP uses configuration policies for data streams that define the data to send to observability back ends. You can use CDP to configure which metrics are collected and how they are processed and transformed within a data stream before being forwarded to their destination.

As a metric source, CDP collects SMF data covering all aspects of z/OS and middleware. For log streaming, CDP supports job logs, the z/OS SYSLOG, z/OS UNIX System Services log files, NetView consolidated audits, NetView and z/OS log (CANZLOG), VSAM, WebSphere Application Server logs, RMF Monitor III reports, and sequential datasets.

As a target, CDP can be configured to send data to IBM Concert for IBM Z and IBM Instana®, and to third-party observability platforms and log management systems. It integrates with Fluentd as a unified logging collector and Apache Kafka for publish/subscribe-based architectures.

The ODP makes metrics from OMEGAMON monitors available to various destinations. ODP supports exporting metrics from monitoring agents for CICS Transaction Server, CICS Transaction Gateway, Db2 for z/OS, IMS, Java virtual machines, IBM MQ for z/OS, Network, Storage, and z/OS. ODP streams metrics in near real time to back ends such as IBM Concert for IBM Z and Instana, integrates with selected observability platforms, and supports outputs to Apache Kafka, OpenTelemetry endpoints, and open-source tools such as Prometheus.

4.2.2 Visibility into transaction traces

For many years, Application Performance Management (APM) solutions used proprietary formats to describe transactional information, along with data-gathering agents that had to be deployed across all environments. This model is problematic in the era of complex hybrid applications that span multiple infrastructure platforms, including the mainframe. If your solution does not support all participants in the application, gaps can occur within the application trace, resulting in a loss of true observability. Also, moving from one vendor's APM solution to another is an expensive exercise that involves replacing agents and potentially reinstrumenting workloads. Most APM solutions also do not provide support across IBM Z, which can lead organizations to perceive the mainframe as a risk within their observability strategy.

IBM Z APM Connect was developed to address this gap and deliver transaction trace information from z/OS to selected APM solutions. With the acquisition of Instana in 2020, IBM invested in adding similar support to this platform, enabling site reliability engineers (SREs) and other users to achieve full end-to-end transaction visibility across their applications, including the mainframe. With the growth of OpenTelemetry as a standard, IBM Z APM Connect was enhanced to capture W3C traceparent information and provide mainframe transaction trace spans in the OpenTelemetry Protocol (OTLP) format. These spans can then be consumed by any observability back end that supports OpenTelemetry.

With trace spans in an observability solution, you can view the overall time that the transaction took to complete and the breakdown of the time that was spent in each step of the transaction flow, along with key contextual information. For example, a span that is generated for a transaction passing through CICS might include details about the region that generated the span, the transaction that was used, the task ID, and the user ID under which the task ran. This visibility helps operations teams quickly isolate where delays in the transaction are most likely occurring and prioritize investigation in those areas.

Figure 4-7 on page 47 shows IBM Instana with CICS spans and attributes.

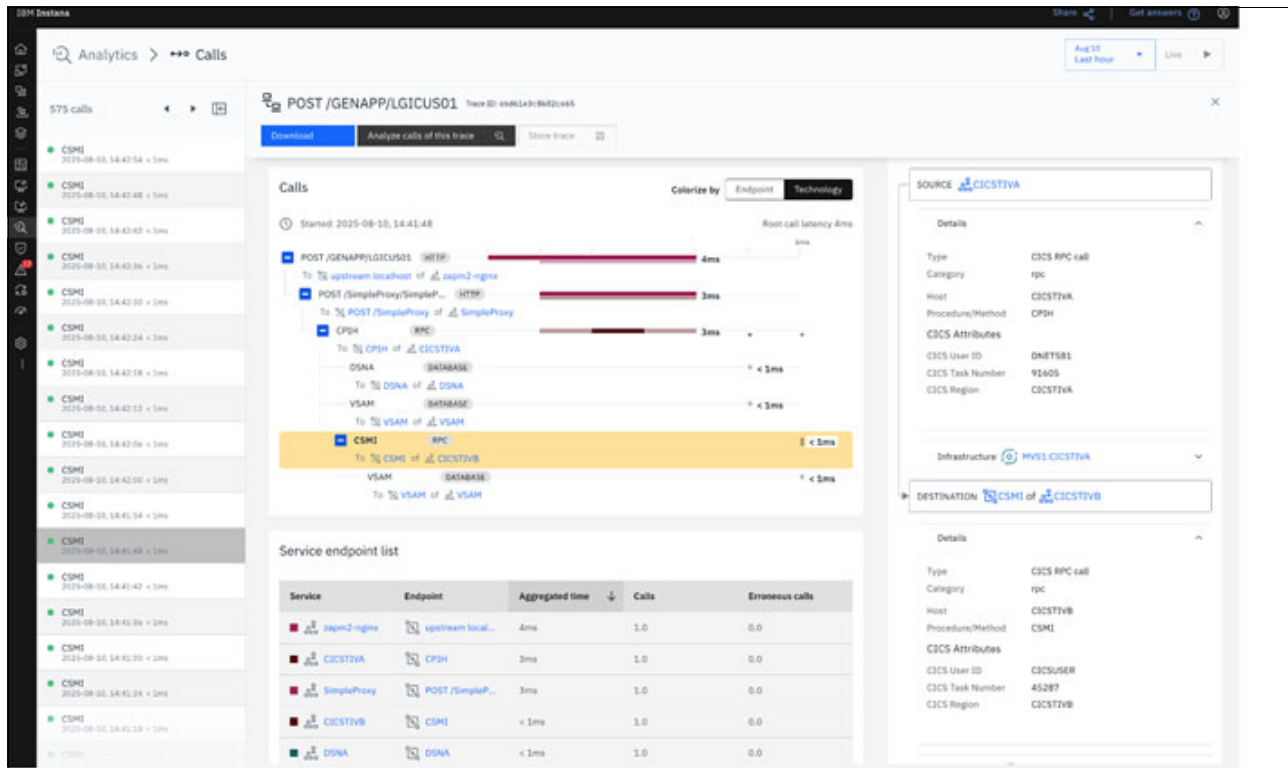


Figure 4-7 Instana with CICS spans and attributes

Many observability solutions are available, both open-source and vendor-supported, but their capabilities can vary. A basic solution might provide only a visualization of received span information, but a complete solution, such as IBM Instana Observability, offers detailed analysis and insights to support the operations team.

Instana is a market-leading, full-stack observability platform for hybrid clouds. With z/OS support, it enables end-to-end observability from mobile through mainframe in a single solution. Instana records 100% of traces without the gaps that are caused by sampling and continuously analyzes span information from all sources to build a comprehensive view of application performance. Instana accepts trace spans from Z APM Connect, OpenTelemetry instrumentation, and its own agent technology. When deviations from expected performance occur, such as an increase in application latency or a sudden drop in transaction volume, an alert is generated and automatically routed to the most appropriate recipient, based on the impacted application or infrastructure component.

By applying AI techniques, Instana performs incident investigation, identifies probable root causes based on the relationships between monitored services, and, where possible, suggests remediation actions to restore services. For mainframe-related issues, the SRE managing the application might involve an expert to resolve the problem. The alerting and context that are provided by Instana make handoffs between team members as seamless as possible.

4.2.3 Augmenting transaction traces with infrastructure metrics

While analyzing application issues, such as increased latency in request response time, you should correlate trace spans with the usage of underlying resources. In Instana, the resources that are associated with trace spans are directly linked to an infrastructure perspective. For example, for a CICS region, Instana visualizes critical performance indicators such as transaction rates, task percentages, CPU usage, I/O rates, and page rates to help you understand the region's health. This data is provided to Instana through the ODP. In addition, OpenTelemetry conforming data can be ingested and visualized in custom dashboards.

4.2.4 Adopting OpenTelemetry

OpenTelemetry plays an essential role in implementing observability solutions for IBM Z and z/OS and in integrating the mainframe into the broader observability ecosystem. The growing adoption of OpenTelemetry as a standard observability framework in enterprises makes it increasingly important for IBM Z and z/OS to support OpenTelemetry. IBM embraces OpenTelemetry as a core element of the observability strategy for IBM Z and LinuxONE. Adopting OpenTelemetry enables IBM Z to become a first-class participant in enterprise observability architectures that are built on OpenTelemetry as a unifying telemetry data layer. The adoption of OpenTelemetry for z/OS aligns with the tiered approach of data producers, aggregators, and consumers.

Data producers

IBM Z APM Connect supports transaction flows across z/OS system software and emits trace spans in OTLP format. In addition, distributed tracing support that uses OpenTelemetry is planned for future releases for CICS, IMS, Db2 for z/OS, and MQ for z/OS. The ODP supports ingesting metrics into back ends by using OTLP. This support maps metrics and resource attributes that are collected by OMEGAMON into OTLP output.

For more information, see [Statement of direction: IBM z/OS middleware products support for OpenTelemetry](#).

Data aggregator

The CDP offers a policy language for filtering, aggregating, and formatting metric data in near real time.

As enterprises revise their observability architectures, the OpenTelemetry Collector is often used for implementing pipelines and serving as a gateway for telemetry data. As the first IBM provided OpenTelemetry Collector distribution, Instana now includes a customized version of the collector.

Data consumer

IBM Concert for IBM Z and Instana support both ingestion and analysis of OTLP data. IBM Concert for IBM Z, with its AI-powered ITOps Management perspective, is designed and built on a modern observability architecture that is centered on structured events. It embraces OpenTelemetry for data ingestion, processing, and distribution. Instana supports seamless integration of OTLP-ingested signals with data that is collected through its agent technology.

Users benefit from the flexibility to instrument their applications and from a rich set of observability features that are built on the data, which support incident analysis and remediation. Trace spans that are sent from z/OS by using OTLP can be viewed alongside spans from services on distributed platforms. Metrics that are ingested by using OTLP can be displayed in custom dashboards and tailored to the needs of users.

4.3 Adopting AI for mainframe operations

This section provides examples that illustrate how AI is adopted, what it does, and how it helps users and the overall business improve within ITOps.

4.3.1 Smart alerts

The principal idea behind smart alerts is to alert only when necessary. Anomaly detection plays a key role by examining a KPI and comparing its current value with the expected value. It is called *smart* because KPI analysis happens continuously and transparently in the background, and users are informed about anomalies automatically.

In contrast to traditional alerting, where a KPI exceeding a static threshold or the presence of a specific message in the z/OS SYSLOG triggers an event, anomaly detection raises an event only in cases of real issues. This approach works often, but false positives can still occur. However, the effort that is required to maintain and constantly adjust static thresholds is reduced to a few edge cases where alerting must occur regardless.

The first example is the AI Data Advisor (AIDA) of the Workload Scheduler (see “AI Data Advisor” on page 59). This capability informs work planners when KPIs that are relevant to a running job, such as the number of completed jobs, exhibit abnormal behavior.

Another example is OMEGAMON AI, which uses a curated set of KPIs to detect abnormal million service units (MSU) consumption, abnormal network traffic (which might indicate a security breach), abnormal garbage collection behavior (which might point to a bug in an updated Java application leaving more memory than before), or unusual CICS transaction rates and response times (which can result from contention on a VSAM data set or the Db2 back end).

IBM Concert for IBM Z, by contrast, observes a larger set of KPIs that are selected by domain experts to apply learning-based anomaly detection and domain-specific expertise for KPI analysis. This process also extends to log data analysis, where anomaly detection and domain knowledge are applied.

4.3.2 Event correlation

To address uncertainty in anomaly detection, IBM Concert for IBM Z aggregates events from diverse KPIs, resources, and external sources (for example, OMEGAMON situations) based on temporal proximity and automatically discovered topological relationships. An algorithm then calculates a confidence score for each event group by applying voting principles and problem pattern recognition. Factors such as multiple anomalies from different perspectives and persistent anomalies over time increase confidence, indicating potentially more impactful or systemic issues.

After forming an event group, IBM Concert for IBM Z displays a dashboard that shows abnormal KPIs and log messages in context. It also provides current state and health dashboards for associated infrastructure resources, transactions, and related elements.

4.3.3 Predictive insights

When the anomaly itself (the single observation that denotes a notable deviation from the expected normal) is considered and the potential impact of that observation is projected, the result is referred to as a *predictive insight*. Various examples fall under this category.

IBM IntelliMagic Vision for z/OS provides health insights in different categories. Although many of these insights are rated based on best practice thresholds that are defined by IBM performance consultants, some of these insights are rated dynamically in correlation with other metrics. Applying statistical methods, such as computing a correlation coefficient between variables, enables the detection of metrics that deviate from normal behavior and require closer examination.

Another predictor is *change detection*. Here, the average of selected metrics over the past 24 hours is compared with the average over the past 30 days. If the current average deviates by more than two (or three) standard deviations from the 30-day average, it is flagged as a notable change in the change detection report, and is marked in yellow or red. For example, if you see a notable change in processor time at an importance level of 0 (system workload) or 1 (high importance), you should investigate the cause to prevent missed service-level objectives or cost impacts.

Although these predictors usually do not require immediate attention, some predictive insights, such as ones that are provided by Workload Scheduler through its What-if analysis, are important because they might have a direct business impact soon. If a critical application must complete before a specific deadline, you should be informed when any of its predecessors are delayed (see 4.3.1, “Smart alerts” on page 49). The work planner can then assess the impact and, if necessary, take immediate action to help ensure timely completion. Even more proactive is the ability to simulate what might happen if a predecessor is delayed or if required input data arrives late. Simulation that is based on historical data helps you evaluate whether your job plan is realistic and optimize the runtime flow to reduce the risk of late job termination.

In general, reviewing predictors regularly is a best practice to stay proactive and reduce the likelihood of outages.

4.3.4 Short interval data correlation

In highly used shared mainframe environments, understanding the interactions among interdependent workload components over time is critical. Various z/OS components and middleware might produce performance data in different formats and at varying intervals, making efficient correlation and analysis difficult due to the lack of standardization. Furthermore, when most analytics products process only aggregated data, even at a 1-minute boundary, short-lived events that might indicate underlying problems might be lost.

IBM z/OS Workload Interaction Correlator (ZWIC) addresses this challenge by providing standardized, synchronized, high-frequency (5-second) data and detailed address space information. This capability enables better correlation and analysis by delivering consistent formats and intervals for performance data.

IBM z/OS Workload Interaction Navigator (ZWIN) builds on ZWIC by analyzing, correlating, and visualizing activities across the entire z/OS software stack. This capability accelerates root cause identification and resolution of operational issues. By examining high-frequency data from ZWIC, ZWIN establishes baselines for typical activity and detects anomalies that signal potential problems.

ZWIN visualization of correlated anomalous activities and insights into interdependencies across z/OS components, middleware, and jobs enables SMEs to quickly narrow their investigation scope and identify the source of issues. The Historical Inspector feature in ZWIN analyzes historical data and identifies recurring or worsening anomaly signatures, enabling proactive diagnosis and mitigation of potential outages or critical situations before they occur.

With this capability, SMEs can quickly determine cause-and-effect relationships across the IBM Z software stack by visualizing workloads across different processor core types and considering job size and priority. Greater visibility into workload interdependencies and interactions helps identify root causes faster and more effectively.

4.3.5 Faster root cause analysis

When an incident is detected, identifying the root cause quickly is critical to plan the necessary actions for remediation or permanent resolution. Often, operators do not know where to start. AI can help detect and prioritize issues so that analysts can focus on what is most critical. In other cases, operators lack insights that are derived from the data that is needed to draw the right conclusions. Generative AI in its different forms can help.

The following sections illustrate how IBM Concert for IBM Z and IntelliMagic Vision for z/OS, which are powered by integration with watsonx Assistant for IBM Z, can expedite this work.

Retrieval augmented generation

When operators do not understand a specific term on the dashboard or are uncertain about the meaning of a message that is issued to the z/OS SYSLOG, they can open the integrated chat and ask a question about the term. IBM watsonx then performs intent analysis, and if the large language model (LLM) determines that documents should be queried, it creates an embedding vector from the user prompt and performs a query against the vector database. The top one, two, or three documents whose vectors most closely match the user prompt are returned and passed along with the user input to the LLM for answering. This method is highly effective for knowledge-based questions because the LLM can provide a concise summary of the most relevant documents. In addition, it can provide links to these sources for further study. Instead of waiting for an available colleague, a chat with the assistant can be the solution.

Looking ahead to agentic AI, retrieval augmented generation (RAG) is often used as one of the tools that provide grounding information for an agent. In this context, *grounding* refers to curated information that is designed to give the agent knowledge, experience, or expertise about the domain it supports. For example, an agent that is skilled in Parallel Sysplex operations must know which hardware components are important, which commands are useful and for what purpose, which performance monitor reports must be analyzed, and more. This grounding information helps the agent reason and plan effectively.

Agentic AI

Agentic AI refers to the multi-step process of reasoning about the user's initial intent, selecting an appropriate action from the available tools or related agents, starting the tool or agent, and evaluating the response. If the response does not match the original intent, this loop can continue for multiple iterations until a limit is reached or a final response is found.

Effective reasoning requires grounding information, which can be obtained, for example, from a vector database by using RAG. Agents cannot use arbitrary tools; they are developed to handle specific tasks, and the agent developer determines which tools are required. For example, if the agent needs a tool to run a z/OS command, that tool is included in the agent's tool set.

The selection of a tool or another agent is performed by the LLM. If the reasoning step suggests displaying a particular system in the sysplex, the LLM attempts to select a tool that can run the corresponding z/OS command. For this process to work, tools, their parameters, and their meanings must be described clearly and concisely. LLMs are effective at selecting the correct tool and populating parameter values that are derived from context. In this example, the parameter is the system name. If the required command is unavailable, the LLM attempts an alternative from the available tools or agents.

After starting the tool (in this example, by running the following z/OS command), the LLM evaluates the response:

```
D XCF,SYSPLEX,systemname
```

If the information in the response matches the user's intent, the result is returned to the user.

Equipped with such tools, products can provide agentic AI capabilities to answer user questions about resource topology, events, and insights from the operating system and subsystems. Products can also help users navigate their ITOps tools by suggesting click-enabled reports that are based on the user's prompt. Operators can interact with the AI through natural language to investigate anomalous events, eliminating the need for manual searches across multiple sources.

4.4 Optimization and availability

Optimization and availability are complementary goals and should be at the core of any modern data center. Optimization, as in business, is a continuous process of improvement. Services must be delivered, but costly resources must also be controlled. Availability is critical in the age of instant media. The cost of an outage is measured in downtime and in reputational impact, which can be devastating and long-lasting. Furthermore, availability is not just a matter of being up or down. Application responsiveness defines the client's perception of effective availability. If response times are too slow, clients take their business elsewhere.

4.4.1 Performance and capacity

The IBM Z platform is a large, shared environment that supports many concurrent and diverse workloads. A unique property of the platform is the interconnected nature of its components. Hardware, hypervisor, operating system, and subsystems work together like an orchestra, delivering the consistent service for which IBM Z is renowned.

Helping ensure that this complex environment continues to run at peak efficiency requires deep insight into both individual components and the collective view of how they work together. Complementing tactical real-time or near real-time solutions such as OMEGAMON and IBM Concert for IBM Z, IBM's strategic planning tool for complete IBM Z insights is IntelliMagic Vision for z/OS.

IntelliMagic Vision for z/OS is built on simplicity. Its foundation is the data that has been at the core of IBM Z logs for decades: SMF records. Every z/OS site automatically collects SMF data with negligible impact. The data is documented, supported as a core component of z/OS, and updated with every release.

A collection of SMF data can be informative but unwieldy. In the past, deep expertise and specialized tools were required to access and interpret this data. For example, Db2 alone produces both statistics (interval-based) and accounting (event-based) records (SMF types 100 - 102). These records contain countless variables that are fed by Db2 Instrumentation Facility Component Identifier (IFCID) trace records. Extracting and interpreting these fields requires effort and expertise.

IntelliMagic Vision for z/OS processes these variables and others with minimal effort, presenting results in consumable charts and tables in a web browser. Although the software can be hosted on-premises, most users choose the cloud-hosted option, which requires only automated data transfer at virtually any chosen interval.

Data can be retained for years, supporting both long-term and short-term strategic analysis and trending. Reports are organized hierarchically, with context-sensitive menus that lead to detailed views. Beyond thousands of stock reports, you can customize or create reports from scratch with a few clicks. Reports can be saved and optionally shared in personal dashboards. All data can also be exported to PDF, PPT, or CSV.

Here are the areas that are covered:

- ▶ IBM Z hardware, including logical and physical chips and memory association
- ▶ Detailed z/OS LPAR components and their interaction and usage
- ▶ Application analysis, including resource consumption (CPU, MIPS, or MSU), response times, and transaction volumes
- ▶ All address spaces and steps
- ▶ SYSPLEX analysis, including Coupling Facility, XCF, and log streams
- ▶ Data storage analysis, including disk, tape, IBM FICON® directors, and datasets
- ▶ Statistics and accounting data for all major subsystems (CICS, Db2, IMS, IBM MQ, TCP/IP, WebSphere, and z/OS Connect)

Beyond ingesting and reporting raw data, IntelliMagic Vision for z/OS enhances the data in many useful ways. Its unique Health Insights feature rates hundreds of metrics by applying known infrastructure limitations and best practices. These known factors do not require lengthy machine learning (ML) because they are derived from expert knowledge in each area. Early warnings, presented as yellow and red shapes that increase in size that is proportionate to severity, often inform staff of approaching issues before they become SLA-impacting incidents.

Figure 4-8 shows a few example reports from IntelliMagic Vision for z/OS.

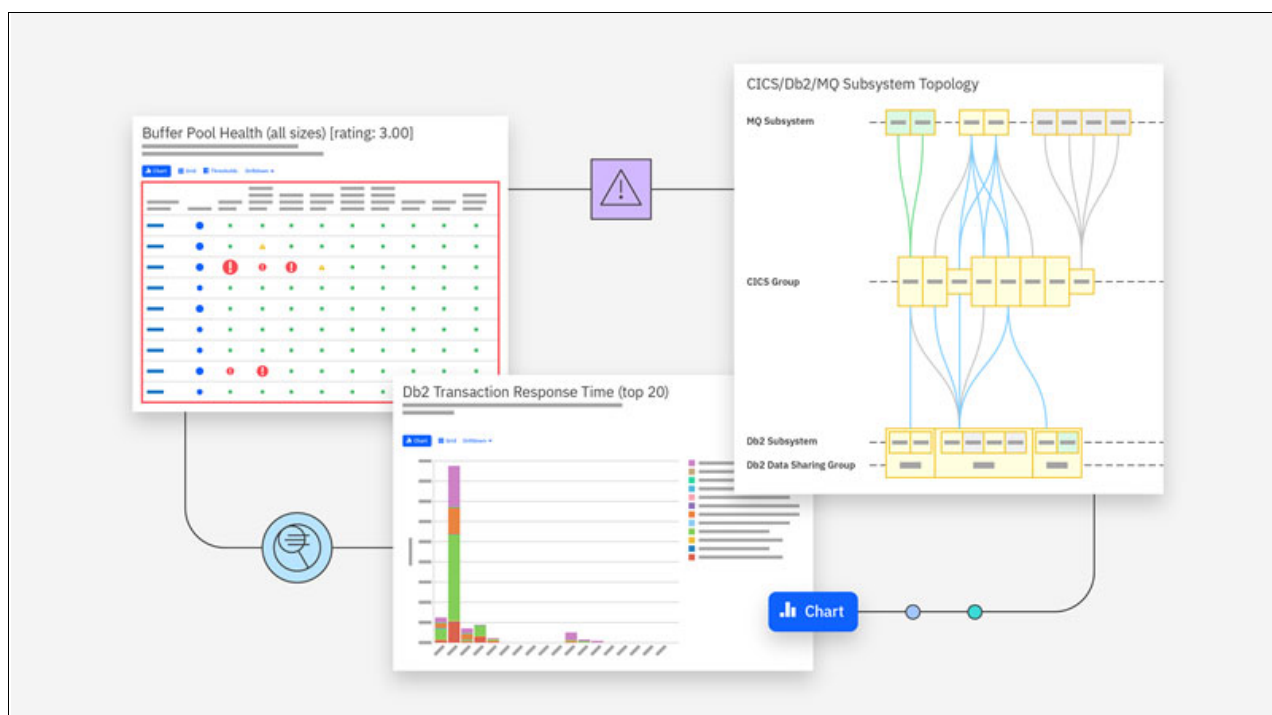


Figure 4-8 IntelliMagic Vision for z/OS report examples

In addition to Health Insights, IntelliMagic Vision for z/OS provides the Change Detection feature. These reports alert analysts to anomalies regardless of whether they strain infrastructure.

IBM Z is highly resilient. For example, an application might unexpectedly double in volume. If IBM Z components absorb the increased load without issue, a health alert might not be triggered. However, detecting the anomaly remains important. Change Detection provides that capability.

SMF automatically reports data by WLM Service Classes and Report Classes, and by transaction and job names. Beyond these labels, many businesses prefer to group workloads into business applications. For example, a bank might have a line of business that is called Loans, with subsets such as Corporate Loans and Personal Loans. An insurance company might have CLAIMS and PAYROLL applications with more subsets. With IntelliMagic Vision for z/OS, you can customize and report on these business structures. This capability supports both service-level reporting and integration with IBM Apptio® to quantify costs for showback.

For data center analysts, IntelliMagic Vision for z/OS includes detailed topology views of their infrastructure. These views are automatically built from the data, with no user input required. Many physical and logical perspectives are available:

- ▶ CEC drawers and chips to LPAR, logical CP, and zIIP core layout
- ▶ LPAR weights and HiperDispatch polarity
- ▶ Channel subsystem to FICON Directors to disk and tape, including individual port addresses

- Cycle and instruction counts and rates, including processor cache and Relative Nest Intensity (RNI)
- Subsystem connections and relationships between CICS regions and transactions, Db2 Data Sharing Groups and members, and IBM MQ Queue Managers

At its core, IntelliMagic Vision for z/OS is designed to be powerful for experienced analysts and approachable for users with less experience. Its intuitive interface guides novice users by providing report explanations and variable names and descriptions. With integration with watsonx Assistant for IBM Z, users can ask questions such as “What are my transaction response times?” or “How busy is LPAR ABC?” The software provides contextual explanations and presents the relevant reports. For seasoned analysts, critical insights such as root cause analysis can be quickly revealed, eliminating outdated methods that require days or weeks to collect and send data, followed by lengthy manual analysis.

It is impractical to list all IntelliMagic Vision for z/OS features here. However, no more comprehensive, strategic solution for the IBM Z performance analyst is known at the time of writing.

4.4.2 Automation: better together

This section focuses on IBM Z System Automation and its underlying automation framework, IBM Z NetView, and explains how they contribute to resiliency. It also explains how you can combine automated operations (AO) with a modern automation framework, such as Ansible, to preserve existing service qualities while introducing flexibility and agility for your DevOps teams.

Highlights

IBM Z System Automation and IBM Z NetView form an enterprise solution for AO. IBM Z NetView provides the automation and application programming framework that enables programs (primarily REXX scripts, but also HLASM, C, or PL/I programs) to trap messages and interact with the system. IBM Z System Automation builds on these capabilities to provide an abstraction layer that you can use to work with automated resources representing single applications or groups of applications, such as started tasks or UNIX System Services processes.

Defining resources and their relationships in a policy database is a major step toward simplification. Before IBM Z System Automation introduced this capability, most automation relied on scripted startup and shutdown. You often maintained hundreds of scripts, which were difficult to test.

AO does not work in isolation. Help ensure that it integrates seamlessly with the surrounding ecosystem. With IBM Z System Automation and IBM Z NetView, standard OpenAPI-conforming REST application programming interfaces (APIs) enable interaction with products running on z/OS or off-platform. You can use this capability through off-platform automation tools such as Ansible or the Zowe CLI.

Simplification also means that you can work on the system without conflicting with automation. This approach is useful when you must perform system maintenance. You can temporarily suspend an application or a group of applications. During this time, IBM Z System Automation ignores messages from the suspended applications and does not perform any automated actions on their behalf. When maintenance is complete, you can resume the application, and IBM Z System Automation regains control.

It is not unusual for z/OS production systems to consist of hundreds of automated resources. Sometimes, the system needs to be started in stages, such as starting all infrastructure (for example, Job Entry Subsystem (JES), TSO, and Db2) first, and then followed by starting the online transaction processing work, such as CICS and IMS, when ready. The opposite direction is often needed: stopping all business workloads and leaving only the bare minimum to perform system maintenance. With IBM Z System Automation, the automation administrator may define run modes, each representing a stage, and associate resources with them. After defining run modes, operators can switch between stages by using a single command.

Standardization is essential in an enterprise environment. IBM Z System Automation encourages the usage of naming conventions and defines an application in the policy database only once, regardless of how many resources of that application exist. For example, the TSO application exists only once in a policy, but you might have many TSO resources, such as one on SYS1, SYS2, and so on. This approach reduces the amount of definition work. If the TSO resource on SYS1 must differ from the TSO resource on SYS2, IBM Z System Automation provides automation symbols that the system can resolve. The definition effort can be further reduced by using application classes to consolidate common definitions at the class level and inherit from that class, rather than reentering the same input repeatedly. For example, think about the many CICS regions you have.

Finally, IBM Z System Automation can create a report about the automation resources in the enterprise, in the many sysplexes and on individual systems. Recently, IBM Z System Automation gained the ability to create reports in JSON format that can be processed off-platform.

IBM Z System Automation Web Console

Modern automation solutions also require modern user interfaces. IBM Z System Automation and NetView provide a browser-based GUI dashboard that operators can use to interact with the system. Through this interface, operators can start or stop resources, maintain schedules, issue commands, and use a consolidated audit, NetView, and z/OS log (CANZLOG) view. This interface helps operators perform their daily tasks without logging on to a 3270 terminal.

Dynamic automated operations

Adding applications to a policy database is possible only with an approved change request. Activating a new automation configuration often requires a regular maintenance interval. However, when a business wants to provision a new started task, such as a CICS region, the operations team cannot immediately respond to the request and provide automation support for it because deploying this change in the automation usually requires one or two weeks of lead time.

Dynamic AO address this need by enabling operators to create resources just in time. The policy contains prepared template definitions, each describing a specific type of resource that should be created dynamically. To create a resource, the operator selects the template and specifies the resource name, the job name, and the system that is assigned to it. All other characteristics are defined in the template.

With this approach, changes to the automation configuration are possible at any time and are still controlled from a change management perspective because the scope of the change is limited by the template, which can be preapproved.

The application department can now create, for example, a CICS region, an IMS message processing region, or an instance of WebSphere Liberty Profile dynamically. To automate this process, they provide the operations team with the job name, the type of application they provisioned, and the system where it was provisioned. The operator can then migrate this application immediately into the active automation configuration.

Ansible automation and recommendations

The section “Dynamic automated operations” on page 56 described the typical approach for provisioning applications traditionally. However, the principle of infrastructure as code is increasingly being adopted in mainframe environments, driving the usage of modern automation frameworks such as Ansible.

Although Ansible is not an ideal tool for AO (consider the many messages that require responses and the failover scenarios that must be addressed), it is an excellent tool for workflows that install and configure software, including on z/OS.

The capability to create resources dynamically can be combined with the provisioning workflow to use the strengths of both automation frameworks. The Ansible Automation Hub provides predefined collections of roles and modules, along with many examples of how to use them. By creating a playbook, customers can build individual workflows that interact with z/OS by using console commands or TSO commands, creating datasets or files, submitting jobs, and more.

A playbook consists of installation and configuration tasks that are essentially all the steps that are required for a system programmer to manually create a started task. Before completing the playbook, you can add a final step to create an IBM Z System Automation dynamic resource for a started task.

Now the playbook can run whenever the application department requires it. It is a fully automated process that includes the operations team, follows the installation’s change management rules, and avoids issues that are caused by a lack of automation support during a temporary period.

Because the question often arises about when to use IBM Z System Automation and when to use Ansible, a few best practices are documented, as shown in Figure 4-9.

Task	IPL / Shutdown	Disaster recovery	Subsystem maintenance	Issuing z/OS commands or scripts for incident remediation	Perform hardware automation
Description	NetView and System Automation (SA) provide robust, reliable, and fast message-based automation to start and stop z/OS workloads during IPL and shutdown without requiring scripts, only based on policy definitions. In addition, Ansible playbooks can trigger IPL via the IBM Z HMC collection and shutdown via the SA REST API.	GDPS is the control point and orchestrator for planned and unplanned outages. Its tight relationship to NetView and SA presumes that latter are used to interface with the hardware and for starting and stopping software. This scenario can be augmented through Ansible, enable Ansible playbooks to initiate planned scenarios via GDPS REST API while orchestrating additional tasks off-platform.	Operations and middleware teams use SA for subsystem maintenance because SA already controls the resources (see tasks on the left) and any action outside of the automation would be interpreted as an exception causing SA to react immediately. In addition, Ansible playbooks can shutdown and recycle middleware workloads via the SA REST API (see also DevOps practices).	Multiple options exist for operators and subject matter experts to execute z/OS commands. Clients want to ensure that all commands and responses appear in a single log accessible by operators. Like existing scripts, Ansible playbooks or rulebooks can serve as new, modern, and reusable automation artifacts within AAP* that can be invoked ad-hoc or automated in response to events. Clients want to always consider message throughput and performance tailored to the specific use case.	SA supports the common tasks for automated hardware operations like activating LPARs, loading control programs, or changing capacity. In addition, Ansible playbooks can perform similar operations using the HMC collection from the AAP. Since GDPS relies on SA's hardware automation, standardizing on SA is recommended from an operations point of view to avoid tracking different logs in case of unforeseen problems.

Figure 4-9 Best practices for automated operations

Figure 4-10 highlights best practices for operations outside the application department that follow a DevSecOps approach with Ansible and IBM Z System Automation.

Task	Dynamic provisioning	Subsystem maintenance	Middleware configuration	SSL certification renewal process	Administrative tasks, e.g., user management
Description	Ansible is best used for dynamic provisioning. Playbooks can leverage existing collections to allow DevOps teams to provision new CICS or IMS regions when needed. Since operations teams must oversee what is running, the playbook can leverage roles from the SA collection from the AAP* to ensure that resources exist and that operators can monitor and operate them.	Ansible playbooks can be invoked to start and stop application and middleware workloads as more and more of these responsibilities are delegated to DevOps teams. However, with SA still in the role as the central point of control for operations, these activities must leverage SA's REST API to inform SA about desired operational status changes.	Ansible is suited for repeated configuration changes, especially if they originate from a central point of control. It promotes configuration as code principles, minimizing errors introduced by manual configuration processes. In addition, Ansible can extend or drive existing automation scripts used for configuration.	Ansible is best used to control the SSL-certificate renewal process as non-z/OS related automation steps might be required here as well, for instance, obtaining a new certificate from a Certificate Authority all while extending or drive existing automation scripts used for z/OS-centric tasks.	Ansible is best used for administrative tasks spanning across platforms, like on/off-boarding a z/OS user in RACF while simultaneously managing them in an enterprise directory. Ansible can drive such administrative tasks either directly or by leveraging existing JCL or automation scripts as a foundation.

* AAP: Red Hat Ansible Automation Platform

Figure 4-10 Best practices for DevSecOps

IBM Geographically Dispersed Parallel Sysplex

Any publication about maintaining resiliency and simplifying mainframe operations is incomplete without mentioning IBM Geographically Dispersed Parallel Sysplex (GDPS).

IBM provides several GDPS solutions that help protect mainframe customers from data loss and outages in disasters. All solutions are based on NetView and IBM Z System Automation, which provide the necessary automation primitives while GDPS delivers the overarching orchestration for the specific solution.

From a resiliency perspective, you can use GDPS to automate the replication of your data within metro distance to another site or across remote distance to another backup region. It also automates the initial program load (IPL), the shutdown of mainframe systems, and the failover process to another site or backup region in a disaster, such as a regional power outage.

At the same time, GDPS simplifies operations. A solution such as a four-site Metro Mirror or Global Mirror disaster recovery (DR) configuration is complex. This solution includes two Parallel Sysplex clusters, each spanning two sites: one in region A and the other in region B. Although data is synchronously mirrored between the sites within a region, it is also asynchronously mirrored across the regions.

Despite this complexity, the operations team does not face it directly. They use the GDPS user interface to monitor the health status of the solution and to handle planned or unplanned outages by using simple commands or Write-To-Operator (WTO) replies. If a disaster occurs and getting online again is the most important goal, you do not want to lose extra time because the operations team must look up guidelines for proper usage.

Cyber resiliency

Traditionally, mainframe teams assumed that the mainframe was protected and physically isolated from external threats. Nothing was expected to go wrong. However, the reality is different. The mainframe is now a central component within a hybrid IT infrastructure and as such is as vulnerable to security breaches, data leaks, and cyberattacks as any other platform in that environment. Therefore, mainframe teams are exploring other strategies to protect critical assets and improve service continuity, which is driven in part by regulations such as the Digital Operational Resiliency Act (DORA).

A data backup strategy is an important element of cyber resiliency. If data is encrypted by ransomware, you must have a backup to restore your assets. But what if the backup itself is already compromised? And how much data is potentially lost after the last valid backup was created?

A combination of storage technology, GDPS, and IBM Z AIOps software can help reduce the risk and limit the impact of a cyberattack. With the IBM DS8000® Safeguarded Copy function, you can create frequent snapshots of your enterprise storage and isolates these backups from physical modification. GDPS Logical Corruption Protection (LCP) then validates each snapshot to determine the most current and valid backup. The validation occurs in a special LCP environment, called the IBM Cyber Vault, which is isolated from the rest of the mainframe infrastructure.

When a cyberattack occurs, the last valid backup can serve as the foundation for recovery. Online transactions (CICS, IMS, or Db2) can be recovered by replaying the transaction logs of the subsystems to further reduce the amount of lost data. IBM Z Backup Resiliency can then identify the datasets that were in use by batch jobs since the last backup was taken and determine which batch jobs must rerun with valid input data after identifying the corresponding backup datasets.

4.4.3 Workload Automation

This section focuses on recent IBM Z Workload Scheduler enhancements that help identify bottlenecks during batch jobs and optimize the plan to help ensure robust and timely workload run times. IBM Z Workload Scheduler is part of the IBM Workload Automation family. It meets all the capabilities that are outlined in 3.4.4, “Batch processing or job scheduling” on page 28 and integrates seamlessly into an enterprise Workload Automation solution that enables customers to automate workloads across the enterprise. Customers can choose whether the enterprise master controller operates from the distributed side or whether all workloads are controlled from z/OS.

AI Data Advisor

AIDA is an anomaly detection engine for Workload Automation. It uses ML algorithms that are trained on historical batch data by using KPIs, such as job duration, the number of completed jobs, the number of waiting jobs by workstation, and more. AIDA can predict the trajectories of KPIs for the current job. If a KPI deviates from expected norms, AIDA immediately alerts work planners so they can prevent problems and delays.

After being informed about an anomaly, a work planner can use the AIDA user interface to investigate the anomalous KPI and drill into other correlated KPIs. For example, when the alert is caused by a smaller number of completed jobs than usual, the work planner can review the trend of waiting jobs that are categorized by the workstation that jobs enables the work planner to identify the possible cause of the delay.

Detecting anomalies as early as possible is a proactive approach that helps reduce operational risks. With AIDA, users can tune the sensitivity for KPIs and select certain days and periods of running jobs where extra tolerances are applied, minimizing the likelihood of false-positive alerts.

Figure 4-11 illustrates how IBM Workload Scheduler AIDA depicts a KPI trend, its expected value range over time, and the anomalies that are detected.

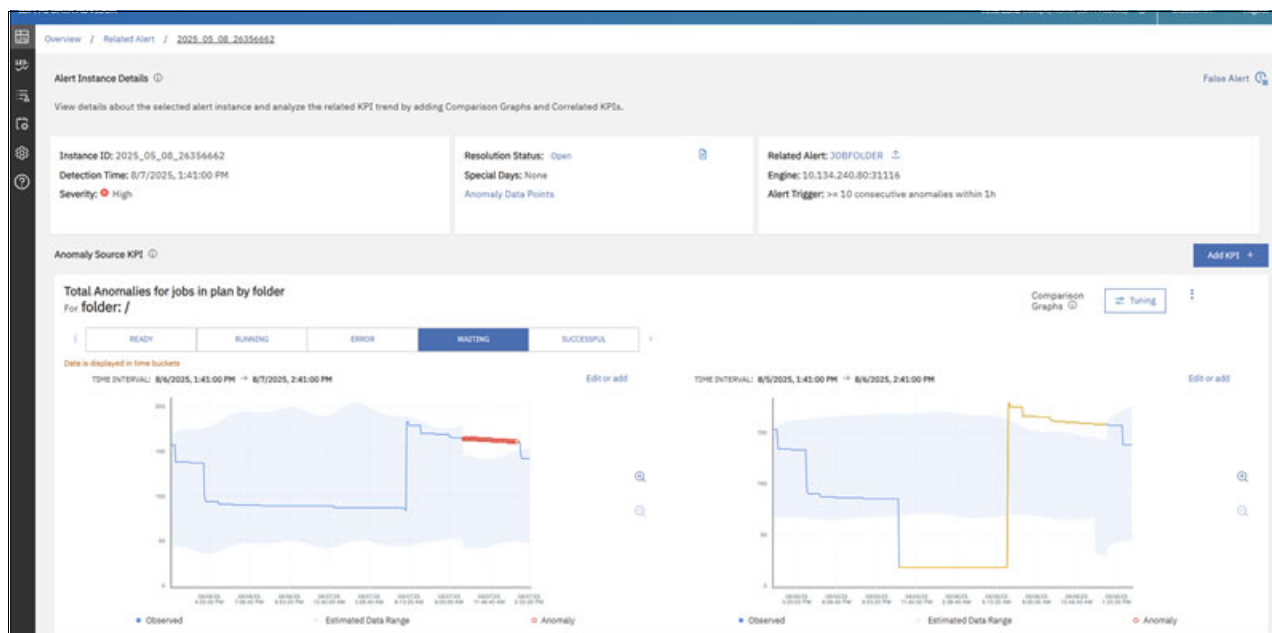


Figure 4-11 AI Data Advisor

More highlights

The following list highlights more features that simplify operations from a Workload Automation perspective:

- ▶ IBM Z Workload Scheduler integrates with observability back ends of your choice by supporting the OTLP for batch jobs. By default, the product supports integration with Instana to send metrics, logs, and traces to a predefined dashboard for monitoring the workload scheduling ecosystem. Users can also choose another back end that can ingest data by using the standardized protocol.
- ▶ Using IBM Z Workload Scheduler, work planners can receive alerts through email and indirectly through a ServiceNow ticket that is opened by an alert condition or through chat notifications that are sent to the collaboration tool of their choice, such as Slack, Microsoft Teams, or Mattermost.
- ▶ IBM Workload Scheduler offers REST APIs that users can use to interact with the scheduler through other software. For example, with the Zowe CLI, users can list jobs in the current plan, change job status, and schedule jobs from a PC terminal window. Similarly, through the integration that is offered by IBM Z ChatOps, users can interact with the scheduler from the chat in watsonx Assistant for IBM Z or from a native collaboration tool such as Slack, Microsoft Teams, or Mattermost.

- ▶ A native file transfer component that is hosted on the distributed side of the Workload Scheduler enables secure file transfers through the agent-to-agent protocol between distributed and mainframe engines without requiring an extra third-party tool.
- ▶ Work planners can publish eligible job streams to a self-service catalog, enabling business users to select on-demand processes from a mobile application or a browser without requiring any Workload Scheduler knowledge.

Where to start

Where and how to begin simplifying mainframe operations depends on your organization's situation, so this question cannot be answered in general terms.

This chapter provides practical tips to help you improve and optimize your existing environment. It uses the capability map that was introduced in Chapter 4, "Simplifying mainframe operations" on page 37 and shown again in Figure 5-1.

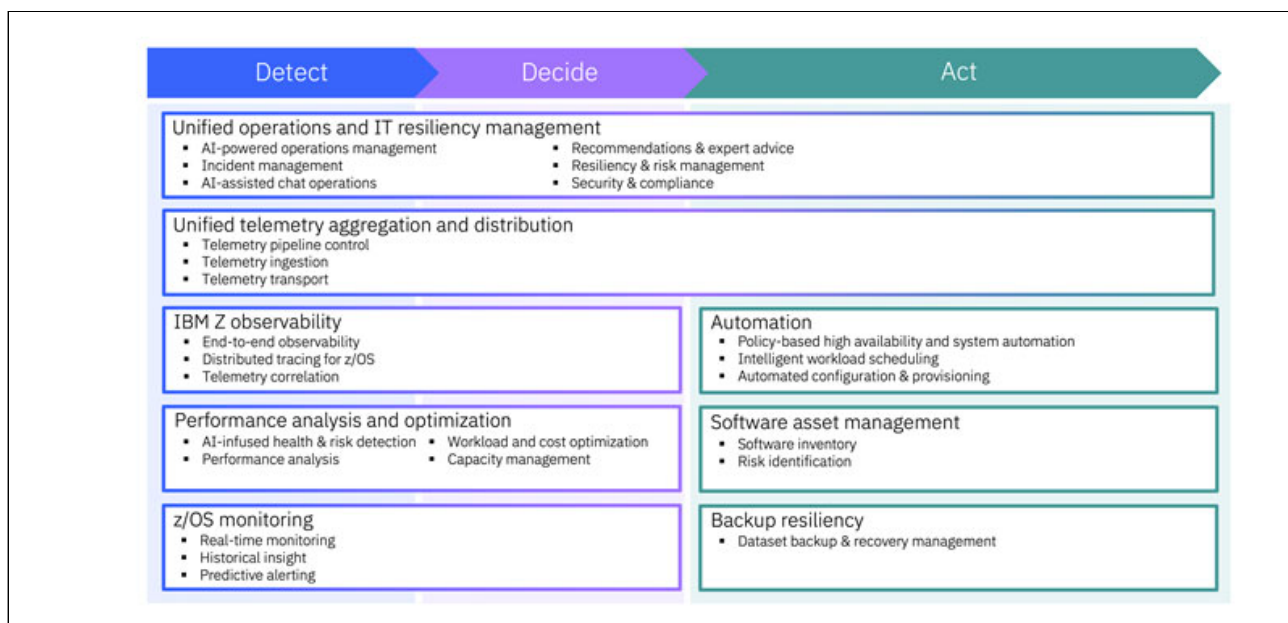


Figure 5-1 IT Operations capabilities

5.1 Establishing a solid monitoring foundation

Every process or action that you perform requires input data as the basis for decision-making. Building on this principle, monitoring is the essential first step toward gaining control and enabling automation. Monitoring is the most fundamental capability for understanding hardware usage, identifying saturation points and capacity needs, measuring transaction durations, evaluating workload performance relative to business importance, and detecting contention. It also helps reduce contention by distributing workloads and data more effectively across the available infrastructure.

IBM OMEGAMON monitors are ideal tools for building this foundation. They provide real-time monitoring and alerting across the entire infrastructure from z/OS to middleware (IBM Customer Information Control System (CICS), IBM Db2, Information Management System (IMS), IBM MQ, and Java monitoring, including z/OS Connect) to network and storage.

The modern Grafana dashboard offers a more intuitive, state-of-the-art experience, replacing the older IBM Tivoli® Enterprise Portal technology. You can also use the interface to tailor dashboards to your organization's specific needs.

OMEGAMON provides fine-tuned anomaly detection for a set of carefully selected metrics that are included in the base. These capabilities are complemented by IBM Concert for IBM Z, which operates further up the stack.

As a result, you can detect and act on unusual million service units (MSU) consumption, network traffic, data set growth, CICS transactions, and Java garbage collection.

The ability of IBM OMEGAMON to stream OpenTelemetry-ready data through OMEGAMON Data Provider (ODP) into IBM Concert for IBM Z or other third-party tools makes it an ideal building block for establishing a broader observability ecosystem.

5.2 Developing a strategy for performance and capacity planning

Real-time monitoring is essential, but you can only react to issues. It is similar to the speedometer in a car. When the speedometer shows that the car is going too fast, the driver eases off the accelerator to slow down. However, the speedometer does not indicate how many miles remain before the next refueling stop or when the vehicle requires maintenance. For that, other tools are needed.

IBM Z IntelliMagic Vision for z/OS Health Insights reports provide a daily view of the infrastructure's health, including IBM Z hardware, the I/O fabric and storage control units, devices, sysplex, z/OS Workload Manager (WLM), middleware, and more. This data helps determine how far you can go or when to consider a capacity upgrade. It also provides change detection reports that uncover hidden patterns by using statistical methods to compare the previous day with the last month. These reports provide performance analysts with hints about where potential problems might exist. With thousands of reports that are included, the product helps performance analysts and capacity planners identify bottlenecks and estimate future capacity needs for business-critical workloads.

The IntelliMagic Vision for z/OS dashboard transformed the way performance analysts work. Its intuitive and visual experience makes performance analysis engaging for the next generation of professionals. Senior experts can create and share dashboards with their teams to explain areas of focus, support onboarding, and broaden the skill base within the profession.

5.3 Establishing standards for automation

Consider how cars were operated in the past. To start the engine, you opened the gas valve first and then turned a crank to initiate ignition so that the engine could run independently. Similarly, starting a z/OS system can be as manual, and it is a cumbersome and lengthy process, often relying on scripts that were created by individuals who might no longer be available.

IBM Z System Automation and the underlying IBM Z NetView provide an ideal solution for performing automated operations (AO) on z/OS. Defining the resources that you want to automate and their relationships to other resources in a policy eliminates the need for scripts so that administrators can focus on what to automate rather than how to automate. Deep integration with z/OS enables immediate monitoring and response to any message. With powerful, practice-oriented concepts, organizations can implement staged initial program load (IPL) and shutdown processes and to automate resources flexibly in a highly dynamic environment while maintaining high availability (HA).

With IBM Z Workload Scheduler, you gain oversight and control of thousands of batch jobs that must run every day. Its ability to plan when and where workloads run and to manage dependencies is essential to help ensure that all relevant batch workloads finish on time. IBM Z Workload Scheduler also assists planners by providing early insights when jobs on the critical path are at risk of finishing late. Whether planners use the powerful what-if analysis to model the consequences of a delay that is caused by a prerequisite job or rely on alerts from AI Data Advisor (AIDA), they can act when they understand the impact. For example, they can hold noncritical jobs to free resources for critical ones. If defined in the plan, Workload Scheduler can expedite delayed jobs on the critical path by resetting them to a service class period with a stronger goal or higher importance.

IBM Z System Automation and IBM Z Workload Scheduler are often used together to better coordinate operations for started tasks and batch jobs. IBM Z System Automation enables the infrastructure that is needed by jobs. For example, setting the state of an IBM Z Workload Scheduler special resource from IBM Z System Automation is an effective way to control the submission of batch jobs that depend on infrastructure software such as CICS or Db2. Conversely, IBM Z Workload Scheduler can delegate operations requests to IBM Z System Automation. For example, when Db2 maintenance is required, many customers run a job stream in IBM Z Workload Scheduler that first delegates the shutdown of Db2 and its dependent applications to IBM Z System Automation before maintenance activities begin. This approach simplifies operations by avoiding duplication of automation knowledge and keeping information in one place.

As the application ecosystem on z/OS becomes more dynamic and might even be controlled from outside the mainframe organization, effective handover of applications to the operations team is critical. For example, you can use Ansible to provision and configure applications on z/OS. By combining Ansible with operations back-end products such as IBM Z System Automation and IBM Z Workload Scheduler, mainframe organizations can be more flexible and responsive to business demands while following best practices and using a single source of automation.

The single source of automation principle is also crucial for disaster recovery (DR) and cyber resiliency. The more operational knowledge is spread across different products and artifacts, the more complex it becomes to orchestrate, maintain, and validate proper failover scenarios.

5.4 Telemetry distribution

To gain a holistic view of your infrastructure, you must distribute metrics, logs, and events to one or more destinations where the data is collected, aggregated, and analyzed. For IntelliMagic Vision for z/OS, the data is periodically prepared and compressed by a batch job and then sent to the IntelliMagic Vision for z/OS instance through secure FTP. For other destinations, such as OMEGAMON AI, IBM Concert for IBM Z, or third-party products, data streaming methods push out new data when it is produced.

While the ODP is used solely for distributing OMEGAMON data, the Common Data Provider (CDP) can distribute any type of System Management Facility (SMF) record, job log, z/OS SYSLOG, NetView CANZLOG, and IMS log. With CDP, you can filter specific data, which helps reduce the amount of data that is sent over the network and stored remotely. It also transforms the data into a format that the destination can process.

As OpenTelemetry adoption grows, telemetry data distribution can also be accomplished by building a telemetry pipeline in which different producers export their signals in a format that follows OpenTelemetry specifications for metrics, logs, and traces.

The distribution mechanisms that you choose depend on the use case. Observability back ends, such as IBM Instana or IBM Concert for IBM Z, support specific mechanisms as part of their evolution and accept OpenTelemetry-based data to enable ingestion from third-party tools.

5.5 Using the unified dashboard for operations

Visibility, control, and automation must work together so that operations teams can detect issues, understand their risk, analyze root causes, and resolve them efficiently. The primary goal of IT Operations (ITOps) is to deliver IT services in accordance with service-level agreements and defined service-level objectives. The faster that issues are detected and the less time operators spend on analysis, the shorter the mean time to resolve.

IBM Concert for IBM Z addresses this need by providing a single, unified user experience across Detect, Decide, and Act. Data is ingested through the CDP, ODP, or an OpenTelemetry pipeline. IBM Concert for IBM Z aggregates data across the IBM Z enterprise and performs anomaly detection. Anomalous observations are rated and grouped into event groups that reduce noise and draw operators' attention to the most critical events. Topology views provide more context to help operators understand the scope of an issue and its blast radius. Context-sensitive dashboards in IBM Concert for IBM Z help you collect detailed data about the system or address space and further isolate the issue. When the root cause is identified, operators can follow the built-in expert recommendations and use the command console to authenticate the system and issue z/OS commands.

With IBM watsonx Assistant for IBM Z integrated into IBM Concert for IBM Z, operators can access extensive information about IBM Z, z/OS, and middleware through a chat interface that is available whenever assistance is needed. Through the same chat, operators can also run predefined automation tasks. The ability to load company-owned documentation into the IBM watsonx Assistant retrieval augmented generation (RAG) vector store helps ensure that operators always have the latest information and guidelines when resolving issues.

5.6 Further advanced steps

There are different paths to continue your journey. The following sections describe the possibilities.

5.6.1 Establishing an observability strategy

The mainframe does not operate in isolation. It is a central and critical platform within a heterogeneous IT infrastructure. Applications across the enterprise access mainframe services to retrieve or update core business data. To understand the performance and availability of these enterprise applications, you need an end-to-end observability strategy.

OpenTelemetry is an open standard for observability that has attracted attention from both the IT industry and clients. While it continues to evolve, it is supported by many observability platforms.

Instana is a full-stack observability platform that supports the OpenTelemetry standard. Therefore, whether today or in the future, the method of data ingestion is less critical with Instana. Instana for Observability on z/OS provides W3C-conforming trace spans of transactions running on the mainframe, and Instana combines these spans into the overall application trace to deliver complete end-to-end visibility. Augmented with metrics from OMEGAMON, Instana provides site reliability engineers (SREs) more infrastructure context to help resolve performance issues faster.

5.6.2 Improving asset management

Mainframe organizations need an inventory of installed software for several reasons:

- ▶ Audit perspective: You can compare licensed software with what is running.
- ▶ Cost perspective: You can determine where software is running and which part of the organization it belongs to, which provides a basis for possible optimizations.
- ▶ Operations perspective: You can identify the software version and determine whether it is approaching or exceeded its support lifecycle.

IBM Z Software Asset Management (IZSAM) provides this visibility. You can use it to quickly identify your IBM Z software inventory and reduce costs while minimizing compliance risk. Together with the z/OS Management Facility, it provides a solution in IBM Concert for IBM Z that helps you determine whether the software level across all z/OS environments is up to date or whether important updates are missing. In this way, it also contributes to the overall resiliency of the infrastructure.

5.6.3 Backup management

The best protection against data loss is a backup. Ideally, you should maintain multiple backups to reduce the risk of having an unusable backup when it is needed.

In highly regulated industries, such as finance and insurance, backups are required to meet compliance rules. IBM Z Backup Resiliency supports mainframe customers by analyzing the data set dependencies of an application and identifying which backups exist or are missing. It also eliminates manual work for setting up the backup process and optimizes resource usage by choosing an appropriate time for backups and avoiding duplicates.

If there is data loss or corruption, having a backup and restore strategy is critical for quickly restoring service. IBM Z Backup Resiliency helps you identify the source of corruption and automate the restoration process. As part of a cyber resiliency strategy, together with IBM Geographically Dispersed Parallel Sysplex (GDPS) Logical Corruption Protection (LCP), and IBM DS8000 Safeguarded Copy, IBM Z Backup Resiliency is a building block that helps protect against the growing threat of ransomware attacks.

Integration architecture

Figure 6-1 shows the IBM Z Artificial Intelligence for IT Operations (AIOps) landscape and the enterprise.

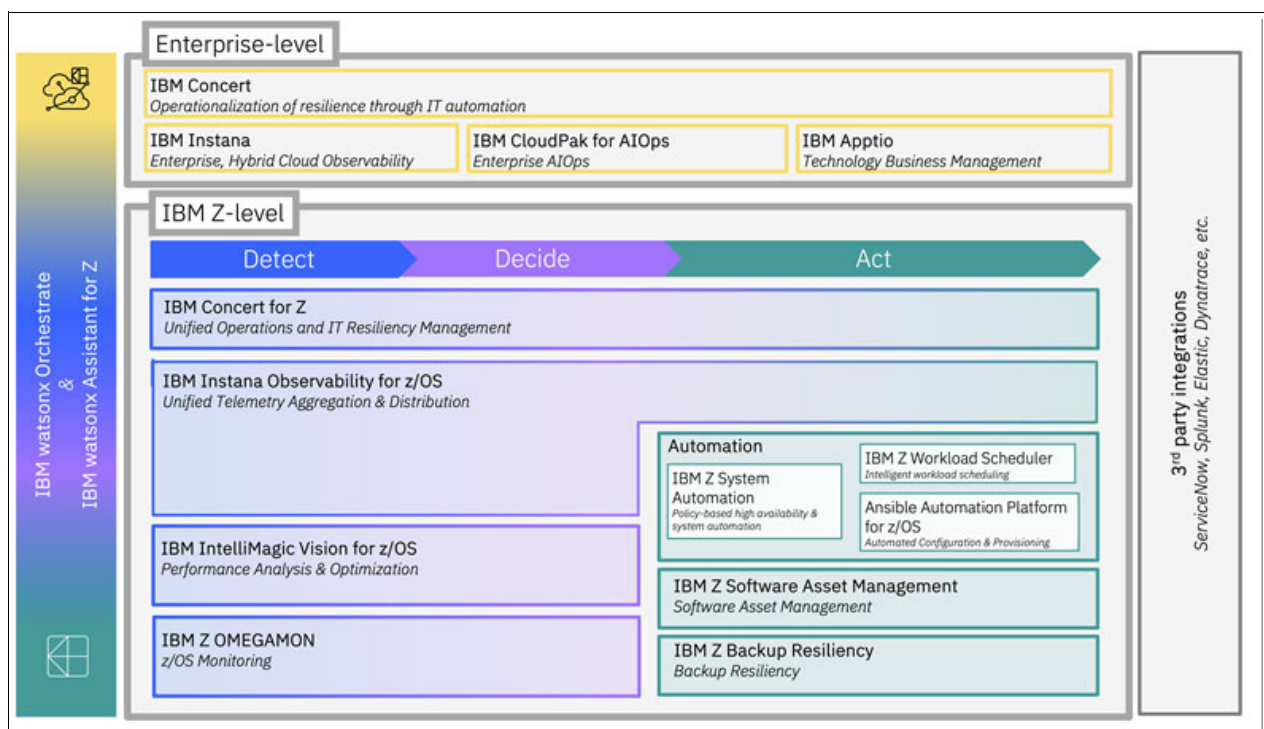


Figure 6-1 IBM Z Artificial Intelligence for IT Operations landscape and the enterprise

At the center of the solution landscape are the products that are described in this IBM Redbooks publication. Up to and including IBM Concert for IBM Z, these solutions focus exclusively on the mainframe, primarily on z/OS.

With the Unified Telemetry Aggregation and Distribution capabilities, which are represented by IBM Instana Observability for z/OS, you can integrate mainframe-generated data with third-party tools in use across the enterprise.

Mainframe data must also integrate with IBM tools for enterprise observability, enterprise AIOps, and technology business management. This integration is not a duplication of effort; it is complementary. For example, IBM Instana reports end-to-end application performance and uses selected OMEGAMON data to annotate transaction hops with infrastructure metrics. IBM Cloud® Pak for AIOps receives alerts through webhook integration whenever IBM Concert for IBM Z detects critical event groups. IBM Z IntelliMagic Vision for z/OS integrates with IBM Apptio to enable better and more granular cost management.

Any solution can use IBM watsonx to enhance its core capabilities to become smarter, increase automation, and help both novice and experienced users through conversational artificial intelligence (AI). In short, watsonx simplifies operations. At the time of writing, IBM Concert for IBM Z users can initiate an AI chat with IBM watsonx Assistant for IBM Z to receive answers. IBM watsonx Assistant for IBM Z responds with references to general z/OS product documentation or company-owned documents. If configured with built-in tools, such as searching for tickets in ServiceNow, these tools can be started directly from the chat without switching to another application.



Conclusion

This IBM Redbooks publication outlined the key challenges that mainframe customers face in IT Operations (ITOps). As IT infrastructure continues to evolve, managing complexity to maintain a resilient and responsive operations environment remains a persistent challenge.

It also described the use cases and specific capabilities that are required for modern operations, which are essential to improving and simplifying IT processes. IBM Z customers follow IT service management (ITSM) processes because these practices have proven successful over many years. These data-driven processes rely on strong monitoring capabilities, which serve as the foundational building block for observability.

You must perform performance and capacity analysis to anticipate future needs and avoid resource bottlenecks or outages that could impact business operations. Policy-driven automated operations (AO) and intelligent job scheduling help you stay efficient as workloads grow. These capabilities also contribute to resiliency and simplification by automating many tasks.

Finally, this IBM Redbooks publication introduced the role of artificial intelligence (AI) in ITOps, defines Artificial Intelligence for IT Operations (AIOps), and explained how the capabilities are organized within the IBM Z AIOps framework built on the pillars Detect, Decide, and Act.

To support simplification and reduce noise, IBM Concert for IBM Z was introduced as the first offering from IBM Z Software that combines all these capabilities. This IBM Redbooks publication also highlights enhancements in the existing IBM Z AIOps product portfolio that strengthen these capabilities and offer evolutionary improvements for further simplification. AI, in various forms, is being adopted across multiple areas. The information that is presented here might soon become outdated as more products incorporate assistive, predictive, and generative AI capabilities.

The following key insights summarize the main takeaways from this IBM Redbooks publication:

- ▶ Simplified and modern operations require multiple capabilities. You cannot focus solely on one area, such as monitoring, while ignoring others. No single tool covers everything.
- ▶ There is strong industry and client demand for end-to-end observability and support for open standards. OpenTelemetry is a rapidly evolving standard for observability, and IBM contributes to its development to enhance observability in the IBM Z space. Although IBM Instana is the preferred choice for enterprise observability, including the mainframe, the commitment to OpenTelemetry means that you can integrate IBM Z data into other third-party back ends.
- ▶ IBM Concert for IBM Z provides a unified solution to detect, group, and prioritize events across the IBM Z enterprise. It helps you quickly perform root cause analysis by using topology views and context-sensitive dashboards and resolve issues efficiently. It integrates seamlessly into incident management processes and includes AI assistance to enhance productivity during incident resolution.
- ▶ Although this IBM Redbooks publication focuses on tools, you must also adjust outdated processes to meet current business demands. People and organizational cultures must evolve to embrace modern processes that are supported by modern tools.
- ▶ AI adoption will continue to help in many ways. The industry is innovating rapidly, and more advancements will follow. However, AI cannot fully replace experts in performance, automation, or z/OS system programming. IBM Z still relies on skilled operators and subject matter experts (SMEs) to make modern operations tools more effective, contributing to the robustness of AIOps in mainframe environments.

Abbreviations and acronyms

AI	artificial intelligence	WLM	Workload Manager
AIDA	AI Data Advisor	WTO	Write-To-Operator
AIOps	Artificial Intelligence for IT Operations	WTOR	Write-To-Operator-with-Response
AO	automated operations	ZWIC	IBM z/OS Workload Interaction Correlator
API	application programming interface	ZWIN	IBM z/OS Workload Interaction Navigator
APM	Application Performance Management		
CDP	Common Data Provider		
ChatOps	chat operations		
CI/CD	continuous integration and continuous deployment		
CICS	Customer Information Control System		
DORA	Digital Operational Resiliency Act		
DR	disaster recovery		
GDPS	Geographically Dispersed Parallel Sysplex		
HA	high availability		
IBM	International Business Machines Corporation		
IMS	Information Management System		
IPL	initial program load		
ITOps	IT Operations		
ITSM	IT service management		
IZSAM	IBM Z Software Asset Management		
JCL	Job Control Language		
JES	Job Entry Subsystem		
KPI	key performance indicator		
LCP	Logical Corruption Protection		
LLM	large language model		
LPAR	logical partition		
MIPS	millions of instructions per second		
ML	machine learning		
MSU	million service units		
ODP	OMEGAMON Data Provider		
OTLP	OpenTelemetry Protocol		
RAG	retrieval augmented generation		
SAF	System Authorization Facility		
SME	subject matter expert		
SMF	System Management Facility		
SRE	site reliability engineer		



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